

Project Data Page:





Table of Contents

Cover Page	1	Nozzle Calcs.: K1	163
Vessel Design Summary:	2	Nozzle Calcs.: M3	168
Nozzle Summary:	5	Nozzle Calcs.: N3	173
Nozzle Schedule:	6		
Bill of Material:	7		
MDMT Summary:	8		
Warnings and Errors:	10		
Input Echo:	11		
XY Coordinate Calculations:	32		
Internal Pressure Calculations:	33		
Element: # 1	35		
Element: # 2	37		
Element: # 3	39		
Element: # 4	41		
Element: # 5	43		
Element: # 6	45		
Element: # 7	47		
Element: # 8	49		
Element: # 9	51		
Element: # 10	53		
Element: # 11	55		
Element: # 12	57		
Element: # 13	59		
Element: # 14	61		
Element: # 15	63		
Element: # 16	65		
Element: # 17	67		
Element: # 18	69		
Element: # 19	71		
Element: # 20	73		
Element: # 21	75		
Element: # 22	77		
Element: # 23	79		
Element: # 24	81		
Element: # 25	83		
Element: # 26	85		
Element: # 27	87		
Element: # 28	89		
Element: # 29	91		
Element: # 30	93		
Shop Pressure Test Results:	95		
External Pressure Calculations:	97		
Element and Detail Weights:	99		
Nozzle Flange MAWP:	102		
Wind Load Calculation:	103		
Earthquake Load Calculation:	106		
Center of Gravity Calculation:	108		
Saddle Calcs: Operating Case:	109		
Saddle Calcs: Test Case:	124		
Nozzle Calcs.: M1	138		
Nozzle Calcs.: N2	143		
Nozzle Calcs.: N1	148		
Nozzle Calcs.: M2	153		
Nozzle Calcs.: K2	158		



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : D:\Yanbu\Bullet\R0\A140NHS10CM001R0.pvdb

Date of Analysis : Sep 8, 2026 9:31am

PV Elite 28, January 2026

[illegible]



Cylinder	55.002	0.00	56.000	2.0000	N/A	No
Cylinder	55.002	0.00	56.000	2.0000	N/A	No
Cylinder	55.002	0.00	56.000	2.0000	N/A	No
Cylinder	55.002	0.00	56.000	2.0000	N/A	No
Cylinder	55.002	0.00	56.000	2.0000	N/A	No
Cylinder	55.002	0.00	56.000	2.0000	N/A	No
Cylinder	55.002	0.00	56.000	2.0000	N/A	No
Sphere	55.001	0.00	57.000	2.0000	N/A	No

Liquid Level: 8880.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.002

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Sphere	0.0	0.0	78.0	74.0	71.5	...	1.00	1.00
Cylinder	3000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	6000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	9000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	12000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	15000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	18000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	21000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	24000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	27000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	30000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	33000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	36000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	39000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	42000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	45000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	48000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	51000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	54000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	57000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	60000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	63000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	66000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	69000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	72000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	74000.0	2000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	77000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	80000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Cylinder	83000.0	3000.0	144.0	144.0	141.0	...	1.00	1.00
Sphere	83000.0	0.0	78.0	74.0	71.5	...	1.00	1.00

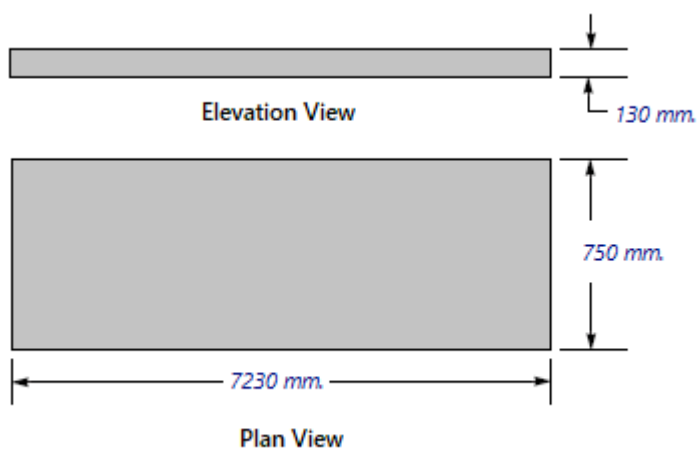
Foundation/Support Summary

Saddle Parameters:

Saddle Width	750.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	4880.000 mm.
Wear Pad Width	900.000 mm.
Wear Pad Thickness	30.000 mm.
Wear Pad Bearing Angle	135.000 deg.
Distance from Saddle to Tangent	8000.000 mm.

Baseplate Length	7230.000 mm.
Baseplate Thickness	130.000 mm.
Baseplate Width	750.000 mm.
Number of Ribs (including outside ribs)	12
Rib Thickness	30.000 mm.
Web Thickness	30.000 mm.
Height of Center Web	576.000 mm.
Number of Bolts in Baseplate	12

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	811179.06	Kgf
Maximum Transverse Saddle Shear Load	62795.34	Kgf
Maximum Longitudinal Saddle Shear Load	251181.36	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	1785124.25	Kgf
Maximum Transverse Saddle Shear Load	8093.55	Kgf
Maximum Longitudinal Saddle Shear Load	5565.59	Kgf

Weights:

Fabricated - Bare w/o removable internals	2594609.8	kg.
Shop Test - Fabricated + water (full)	7072406.5	kg.
Shipping - Fab. + rem. intls.+ shipping app.	2617999.8	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	2650499.8	kg.
Empty - Fab. + intls. + details + wghts.	2650499.8	kg.
Operating - Empty + operating liquid (no ca)	2657415.5	kg.



Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
M1	55	...	...	...	OK	Passed
N2	56	...	...	OK	OK	Passed
N1	55	...	...	OK	OK	Passed
M2	55	...	...	...	OK	Passed
K2	56	...	...	OK	OK	Passed
K1	56	...	...	OK	OK	Passed
M3	55	...	...	...	OK	Passed
N3	56	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 55 Nozzle : N1
 Minimum MAWP Shells/Flanges : 55 Element :
 Minimum MAPnc Shells/Flanges : 62 Element :

Computed Vessel M.A.W.P. : 55 bars

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Inside Radius mm.
30	M1	4500.000	90.000	307.800
40	N2	8499.980	90.000	535.400
40	N1	6499.975	90.000	256.000
160	M2	43500.004	90.000	307.800
270	K2	74499.984	90.000	27.400
270	K1	76499.984	90.000	27.400
280	M3	78499.977	90.000	307.800
290	N3	81499.977	270.000	40.100

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2	2.000 in	Actual	WNF	3.339	17.000	...	...	707.30	600
K1	2.000 in	Actual	WNF	3.339	17.000	...	...	707.30	600
N3	3.000 in	Actual	None	4.575	20.000	...	...	394.97	...
N1	20.000 in	Actual	WNF	20.945	12.000	...	...	607.97	600
M1	24.000 in	Actual	WNF	25.260	16.000	...	...	598.89	600
M2	24.000 in	Actual	WNF	25.260	16.000	...	...	598.89	600
M3	24.000 in	Actual	WNF	25.260	16.000	...	...	598.89	600
N2	42.000 in	Actual	WNF	43.732	22.000	...	...	565.64	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2	SA-350 LF2	144.000	10.000	...	...	...
K1	SA-350 LF2	144.000	10.000	...	...	...
N3	SA-350 LF2	144.000	10.000	...	...	...
N1	SA-350 LF2	144.000	10.000	...	...	...
M1	SA-350 LF2	31.000	10.000	...	...	...
M2	SA-350 LF2	31.000	10.000	...	...	...
M3	SA-350 LF2	31.000	10.000	...	...	...
N2	SA-350 LF2	144.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
K2	74499.984	90.0	562.85	0.00	Node: 270
K1	76499.984	90.0	562.85	0.00	Node: 270
N3	81499.977	270.0	250.00	0.00	Node: 290
N1	6499.975	90.0	445.50	0.00	Node: 40
M1	4500.000	90.0	432.80	0.00	Node: 30
M2	43500.004	90.0	432.80	0.00	Node: 160
M3	78499.977	90.0	432.80	0.00	Node: 280
N2	8499.979	90.0	356.60	0.00	Node: 40

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 78.0mm. THK X 8070.0mm. ID X 0.0mm.	SA-516 70
27	CYLINDER: 144.0mm. THK X 8000.0mm. ID X 3000.0mm.	SA-516 70
1	CYLINDER: 144.0mm. THK X 8000.0mm. ID X 2000.0mm.	SA-516 70
2	PLATFORM: 1200mm X 6000mm WIDE	
3	LADDER: 9660.0mm	...
4	SADDLE: 750.0mm X 120 DEG	SA-516 70
1	PLATFORM: 1200mm X 3000mm WIDE	
3	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	NAMEPLATE	...

Description	Notes	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
	[9]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
	[8]	!	0		Yes			
M1	[1]	!	8	-2	Yes	0.803	16.000	1.00
Nozzle Flg	[5]	A	-18	-46	Yes			
N2	[1]	!	-46		Yes			
Nozzle Flg	[5]	!	-46	-73	Yes			
N1	[1]	!	3	-3	Yes	0.868	12.000	1.00
Nozzle Flg	[5]	A	-18	-46	Yes			
M2	[1]	!	8	-2	Yes	0.803	16.000	1.00
Nozzle Flg	[5]	A	-18	-46	Yes			
K2	[1]	!	-46		Yes			
Nozzle Flg	[5]	A	-18	-46	Yes			
K1	[1]	!	-46		Yes			
Nozzle Flg	[5]	A	-18	-46	Yes			
M3	[1]	!	8	-2	Yes	0.803	16.000	1.00
Nozzle Flg	[5]	A	-18	-46	Yes			
N3	[1]	!	-46		Yes			
Warmest MDMT:			8	-2				

Notes:

- [1] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3



- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	55 bars
Design Internal Temperature	85.0 °C
Projection of Nozzle from Vessel Top	0 mm.
Projection of Nozzle from Vessel Bottom	0 mm.
Minimum Design Metal Temperature	0.0 °C
Type of Construction	Welded
Special Service	None
Use Higher Longitudinal Stresses (Flag)	N
Select t for Internal Pressure (Flag)	N
Select t for External Pressure (Flag)	N
Select t for Axial Stress (Flag)	N
Select Location for Stiff. Rings (Flag)	N
Consider Vortex Shedding	N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	500	mm.
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh 0	mm.
Distance Upwind of Crest	Lh 0	mm.
Distance from Crest to the Vessel	x 0	mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h 0.000	
Amplification Factor	Ap 2.500	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR] 0.000	
Component strength factor	[Rpo] 0.000	
Overstrength factor	[Omega0] 0.000	
Response Acceleration Factor	[SMS] 0.326	
Response Acceleration Factor	[SM1] 0.139	
Building Importance Factor	[Ie] 1.000	
Site Latitude	0.000	deg
Site Longitude	0.000	deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/30
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	
Distance "FROM" to "TO"	0 mm.
Inside Diameter	8070 mm.
Element Thickness	74 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	78 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C



Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	175 N./mm ²
Allowable Stress, Operating	161.14 N./mm ²
Allowable Stress, Hydrotest	247 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm.
Yield Stress, Operating	242 N./mm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8070 mm.
Liquid Density	0.17970E-05 kg./cm ³

Element From Node	10
Detail Type	Weight
Detail ID	cut back-1
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	7830 Kg
Offset from Element Centerline	0 mm.

Element#	2/30
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	297960 cm ²
Platform Weight	10688 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm ²
Platform Width	6000 mm.
Platform Height	1200 mm.
Platform Clearance or End Offset	200 mm.
Platform Force Coefficient	2.0
Ladder Layout Angle	0.0



Ladder Start Elevation 0 mm.
 Ladder End Elevation 9660 mm.
 Unit Weight of Ladder 0.05 Kgf/mm.
 Platform Length (top head platform) 10000 mm.

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid: 20
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 8000 mm.
 Liquid Density 0.17970E-05 kg./cm³

Element From Node 20
 Detail Type Weight
 Detail ID MH Davit
 Dist. from "FROM" Node / Offset dist 3000 mm.
 Miscellaneous Weight 100 Kgf
 Offset from Element Centerline 4500 mm.

Element From Node 20
 Detail Type Weight
 Detail ID PL clip
 Dist. from "FROM" Node / Offset dist 3000 mm.
 Miscellaneous Weight 650 Kgf
 Offset from Element Centerline 4209 mm.

Element# 3/30
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 8000 mm.
 Element Thickness 144 mm.
 Internal Corrosion Allowance 2 mm.
 Nominal Thickness 144 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 55 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 30
 Detail Type Liquid
 Detail ID Liquid: 30
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 8880 mm.
 Liquid Density 0.17970E-05 kg./cm³

Element From Node 30
 Detail Type Nozzle
 Detail ID M1
 Dist. from "FROM" Node / Offset dist 1500 mm.
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 600
 Layout Angle 90.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 0 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 30



Detail Type	Weight
Detail ID	INT Ladder-1
Dist. from "FROM" Node / Offset dist	1500 mm.
Miscellaneous Weight	500 Kg
Offset from Element Centerline	0 mm.

Element#	4/30
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Saddle-1
Dist. from "FROM" Node / Offset dist	2000 mm.
Width of Saddle	750 mm.
Height of Saddle at Bottom	4880 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	900 mm.
Thickness of Wear Plate	30 mm.
Contact Angle, Wear Plate (degrees)	135.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	N2
Dist. from "FROM" Node / Offset dist	2500 mm.
Nozzle Diameter	42 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	N1



Dist. from "FROM" Node / Offset dist	499.98 mm.
Nozzle Diameter	20 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	40
Detail Type	Weight
Detail ID	FP-1
Dist. from "FROM" Node / Offset dist	2000 mm.
Miscellaneous Weight	7000 Kgf
Offset from Element Centerline	4000 mm.

Element From Node	40
Detail Type	Weight
Detail ID	Fire Cleat-1
Dist. from "FROM" Node / Offset dist	3000 mm.
Miscellaneous Weight	2000 Kgf
Offset from Element Centerline	0 mm.

Element From Node	40
Detail Type	Weight
Detail ID	CV-1
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	3500 Kgf
Offset from Element Centerline	4500 mm.

Element From Node	40
Detail Type	Weight
Detail ID	Valve
Dist. from "FROM" Node / Offset dist	500 mm.
Miscellaneous Weight	500 Kgf
Offset from Element Centerline	4500 mm.

Element#	5/30
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm³



Element#	6/30
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm ³
Element From Node	60
Detail Type	Weight
Detail ID	misc-1
Dist. from "FROM" Node / Offset dist	-0.02442 mm.
Miscellaneous Weight	5000 Kgf
Offset from Element Centerline	0 mm.
Element From Node	60
Detail Type	Weight
Detail ID	Fire cleat-1
Dist. from "FROM" Node / Offset dist	-0.02442 mm.
Miscellaneous Weight	1700 Kgf
Offset from Element Centerline	0 mm.

Element#	7/30
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.



Liquid Density 0.17970E-05 kg./cm³

Element# 8/30
 Element From Node 80
 Element To Node 90
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 8000 mm.
 Element Thickness 144 mm.
 Internal Corrosion Allowance 2 mm.
 Nominal Thickness 144 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 55 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 80
 Detail Type Liquid
 Detail ID Liquid: 80
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 8880 mm.
 Liquid Density 0.17970E-05 kg./cm³

Element# 9/30
 Element From Node 90
 Element To Node 100
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 8000 mm.
 Element Thickness 144 mm.
 Internal Corrosion Allowance 2 mm.
 Nominal Thickness 144 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 55 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 90
 Detail Type Liquid
 Detail ID Liquid: 100
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 8880 mm.
 Liquid Density 0.17970E-05 kg./cm³

Element# 10/30
 Element From Node 100
 Element To Node 110
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.



Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid: 110	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8880	mm.
Liquid Density	0.17970E-05	kg./cm ³

Element#	11/30	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	110	
Detail Type	Liquid	
Detail ID	Liquid: 120	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8880	mm.
Liquid Density	0.17970E-05	kg./cm ³

Element#	12/30	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	



Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 120
 Detail Type Saddle
 Detail ID New Saddle
 Dist. from "FROM" Node / Offset dist 333 mm.
 Width of Saddle 750 mm.
 Height of Saddle at Bottom 4768 mm.
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm.
 Width of Wear Plate 900 mm.
 Thickness of Wear Plate 30 mm.
 Contact Angle, Wear Plate (degrees) 135.0
 Friction coefficient 0.1
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm.
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 mm.

Element From Node 120
 Detail Type Weight
 Detail ID FP2
 Dist. from "FROM" Node / Offset dist 333 mm.
 Miscellaneous Weight 7000 Kgf
 Offset from Element Centerline 4000 mm.

Element# 13/30
 Element From Node 130
 Element To Node 140
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 8000 mm.
 Element Thickness 144 mm.
 Internal Corrosion Allowance 2 mm.
 Nominal Thickness 144 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 55 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 130
 Detail Type Liquid
 Detail ID Liquid: 130
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 8880 mm.
 Liquid Density 0.17970E-05 kg./cm³

Element# 14/30
 Element From Node 140
 Element To Node 150
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 8000 mm.
 Element Thickness 144 mm.



Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element#	15/30	
Element From Node	150	
Element To Node	160	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element#	16/30	
Element From Node	160	
Element To Node	170	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	160	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	129960	cm²
Platform Weight	2013.8	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm²
Platform Width	3000	mm.



Platform Height	1200	mm.
Platform Clearance or End Offset	200	mm.
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	9660	mm.
Unit Weight of Ladder	0.05	Kgf/mm.
Platform Length (top head platform)	3000	mm.

Element From Node	160	
Detail Type	Liquid	
Detail ID	Liquid: 140	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8880	mm.
Liquid Density	0.17970E-05	kg./cm ³

Element From Node	160	
Detail Type	Nozzle	
Detail ID	M2	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

Element From Node	160	
Detail Type	Weight	
Detail ID	PL clip-3	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	4200	mm.

Element From Node	160	
Detail Type	Weight	
Detail ID	MH Devit	
Dist. from "FROM" Node / Offset dist	1500	mm.
Miscellaneous Weight	100	Kgf
Offset from Element Centerline	4500	mm.

Element From Node	160	
Detail Type	Weight	
Detail ID	INT Ladder-2	
Dist. from "FROM" Node / Offset dist	1500	mm.
Miscellaneous Weight	500	Kgf
Offset from Element Centerline	0	mm.

Element#	17/30	
Element From Node	170	
Element To Node	180	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	



Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element# 18/30
 Element From Node 180
 Element To Node 190
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 8000 mm.
 Element Thickness 144 mm.
 Internal Corrosion Allowance 2 mm.
 Nominal Thickness 144 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 55 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 180
 Detail Type Liquid
 Detail ID Liquid: 160
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 8880 mm.
 Liquid Density 0.17970E-05 kg./cm³

Element# 19/30
 Element From Node 190
 Element To Node 200
 Element Type Cylinder
 Description
 Distance "FROM" to "TO" 3000 mm.
 Inside Diameter 8000 mm.
 Element Thickness 144 mm.
 Internal Corrosion Allowance 2 mm.
 Nominal Thickness 144 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 55 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 190
 Detail Type Saddle
 Detail ID New Saddle
 Dist. from "FROM" Node / Offset dist 1666 mm.
 Width of Saddle 750 mm.
 Height of Saddle at Bottom 4656 mm.
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm.
 Width of Wear Plate 900 mm.
 Thickness of Wear Plate 30 mm.
 Contact Angle, Wear Plate (degrees) 135.0
 Friction coefficient 0.1
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm.



Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	190	
Detail Type	Liquid	
Detail ID	Liquid: 170	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8880	mm.
Liquid Density	0.17970E-05	kg./cm ³
Element From Node	190	
Detail Type	Weight	
Detail ID	FP3	
Dist. from "FROM" Node / Offset dist	1666	mm.
Miscellaneous Weight	7000	Kgf
Offset from Element Centerline	4000	mm.

Element#	20/30	
Element From Node	200	
Element To Node	210	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	200	
Detail Type	Liquid	
Detail ID	Liquid: 180	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8880	mm.
Liquid Density	0.17970E-05	kg./cm ³

Element#	21/30	
Element From Node	210	
Element To Node	220	
Element Type	Cylinder	
Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +IT	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element From Node	210
Detail Type	Liquid
Detail ID	Liquid: 190
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm ³

Element#	22/30
Element From Node	220
Element To Node	230
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	220
Detail Type	Liquid
Detail ID	Liquid: 210
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm ³

Element#	23/30
Element From Node	230
Element To Node	240
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	230
Detail Type	Liquid
Detail ID	Liquid: 220
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm ³



Element#	24/30
Element From Node	240
Element To Node	250
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	240
Detail Type	Liquid
Detail ID	Liquid: 230
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm ³
Element From Node	240
Detail Type	Weight
Detail ID	misc 2
Dist. from "FROM" Node / Offset dist	3000 mm.
Miscellaneous Weight	5000 Kg
Offset from Element Centerline	0 mm.
Element From Node	240
Detail Type	Weight
Detail ID	Fire cleat-2
Dist. from "FROM" Node / Offset dist	3000 mm.
Miscellaneous Weight	1700 Kg
Offset from Element Centerline	0 mm.

Element#	25/30
Element From Node	250
Element To Node	260
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	250
Detail Type	Liquid
Detail ID	Liquid: 240
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.



Liquid Density 0.17970E-05 kg./cm³

Element#	26/30
Element From Node	260
Element To Node	270
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	2000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	260
Detail Type	Liquid
Detail ID	Liquid: 250
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm ³

Element#	27/30
Element From Node	270
Element To Node	280
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	270
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	1000 mm.
Width of Saddle	750 mm.
Height of Saddle at Bottom	4545 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	900 mm.
Thickness of Wear Plate	30 mm.
Contact Angle, Wear Plate (degrees)	135.0
Friction coefficient	0.1
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0



Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	270	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	297960	cm ²
Platform Weight	10688	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm ²
Platform Width	6000	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	200	mm.
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	9660	mm.
Unit Weight of Ladder	0.05	Kgf/mm.
Platform Length (top head platform)	10000	mm.
Element From Node	270	
Detail Type	Liquid	
Detail ID	Liquid: 260	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8880	mm.
Liquid Density	0.17970E-05	kg./cm ³
Element From Node	270	
Detail Type	Nozzle	
Detail ID	K2	
Dist. from "FROM" Node / Offset dist	500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	270	
Detail Type	Nozzle	
Detail ID	K1	
Dist. from "FROM" Node / Offset dist	2500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	270	
Detail Type	Weight	
Detail ID	PL clip-2	
Dist. from "FROM" Node / Offset dist	3000	mm.
Miscellaneous Weight	650	Kgf
Offset from Element Centerline	4209	mm.
Element From Node	270	
Detail Type	Weight	
Detail ID	FP4	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	7000	Kgf
Offset from Element Centerline	4000	mm.



Element From Node	270
Detail Type	Weight
Detail ID	Fire cleat-2
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	2000 Kgf
Offset from Element Centerline	0 mm.

Element#	28/30
Element From Node	280
Element To Node	290
Element Type	Cylinder
Description	
Distance "FROM" to "TO"	3000 mm.
Inside Diameter	8000 mm.
Element Thickness	144 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	144 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	55 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	280
Detail Type	Liquid
Detail ID	Liquid: 270
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	8880 mm.
Liquid Density	0.17970E-05 kg./cm³

Element From Node	280
Detail Type	Nozzle
Detail ID	M3
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	280
Detail Type	Weight
Detail ID	MH davit
Dist. from "FROM" Node / Offset dist	2000 mm.
Miscellaneous Weight	50 Kgf
Offset from Element Centerline	4500 mm.

Element From Node	280
Detail Type	Weight
Detail ID	INT Ladder-3
Dist. from "FROM" Node / Offset dist	1500 mm.
Miscellaneous Weight	500 Kgf
Offset from Element Centerline	0 mm.

Element#	29/30
Element From Node	290
Element To Node	300
Element Type	Cylinder



Description		
Distance "FROM" to "TO"	3000	mm.
Inside Diameter	8000	mm.
Element Thickness	144	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	144	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	290	
Detail Type	Liquid	
Detail ID	Liquid: 280	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8880	mm.
Liquid Density	0.17970E-05	kg./cm ³
Element From Node	290	
Detail Type	Nozzle	
Detail ID	N3	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-350 LF2	+IT

Element#	30/30	
Element From Node	300	
Element To Node	310	
Element Type	Spherical	
Description		
Distance "FROM" to "TO"	0	mm.
Inside Diameter	8070	mm.
Element Thickness	74	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	78	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	55	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	300	
Detail Type	Liquid	
Detail ID	Liquid: 290	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	8070	mm.
Liquid Density	0.17970E-05	kg./cm ³
Element From Node	300	
Detail Type	Weight	
Detail ID	cut back 2	



Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	7830	Kgf
Offset from Element Centerline	0	mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
10	20	...	...	...	...
20	30	3000	...	3000	...
30	40	6000	...	3000	...
40	50	9000	...	3000	...
50	60	12000	...	3000	...
60	70	15000	...	3000	...
70	80	18000	...	3000	...
80	90	21000	...	3000	...
90	100	24000	...	3000	...
100	110	27000	...	3000	...
110	120	30000	...	3000	...
120	130	33000	...	3000	...
130	140	36000	...	3000	...
140	150	39000	...	3000	...
150	160	42000	...	3000	...
160	170	45000	...	3000	...
170	180	48000	...	3000	...
180	190	51000	...	3000	...
190	200	54000	...	3000	...
200	210	57000	...	3000	...
210	220	60000	...	3000	...
220	230	63000	...	3000	...
230	240	66000	...	3000	...
240	250	69000	...	3000	...
250	260	72000	...	3000	...
260	270	74000	...	2000	...
270	280	77000	...	3000	...
280	290	80000	...	3000	...
290	300	83000	...	3000	...
300	310	83000	...	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
10	20	55.001	78	2	8070	161.14	1
20	30	55.001	144	2	8000	161.1	2
30	40	55.002	144	2	8000	161.1	3
40	50	55.002	144	2	8000	161.1	4
50	60	55.002	144	2	8000	161.1	5
60	70	55.002	144	2	8000	161.1	6
70	80	55.002	144	2	8000	161.1	7
80	90	55.002	144	2	8000	161.1	8
90	100	55.002	144	2	8000	161.1	9
100	110	55.002	144	2	8000	161.1	10
110	120	55.002	144	2	8000	161.1	11
120	130	55	144	2	8000	161.1	12
130	140	55.002	144	2	8000	161.1	13
140	150	55	144	2	8000	161.1	14
150	160	55	144	2	8000	161.1	15
160	170	55.002	144	2	8000	161.1	16
170	180	55	144	2	8000	161.1	17
180	190	55.002	144	2	8000	161.1	18
190	200	55.002	144	2	8000	161.1	19
200	210	55.002	144	2	8000	161.1	20
210	220	55.002	144	2	8000	161.1	21
220	230	55.002	144	2	8000	161.1	22
230	240	55.002	144	2	8000	161.1	23
240	250	55.002	144	2	8000	161.1	24
250	260	55.002	144	2	8000	161.1	25
260	270	55.002	144	2	8000	161.1	26
270	280	55.002	144	2	8000	161.1	27
280	290	55.002	144	2	8000	161.1	28
290	300	55.002	144	2	8000	161.1	29
300	310	55.001	78	2	8070	161.1	30

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
10	20	55	56.9678	63.602	74	71.4917	1
20	30	55	56.1664	61.8889	144	141	2
30	40	55	56.1662	61.8889	144	141.001	3
40	50	55	56.1662	61.8889	144	141.001	4
50	60	55	56.1662	61.8889	144	141.001	5
60	70	55	56.1662	61.8889	144	141.001	6
70	80	55	56.1662	61.8889	144	141.001	7
80	90	55	56.1662	61.8889	144	141.001	8
90	100	55	56.1662	61.8889	144	141.001	9
100	110	55	56.1662	61.8889	144	141.001	10
110	120	55	56.1662	61.8889	144	141.001	11
120	130	55	56.1678	61.8889	144	140.996	12
130	140	55	56.1662	61.8889	144	141.001	13
140	150	55	56.1678	61.8889	144	140.996	14
150	160	55	56.1678	61.8889	144	140.996	15
160	170	55	56.1662	61.8889	144	141.001	16
170	180	55	56.1678	61.8889	144	140.996	17
180	190	55	56.1662	61.8889	144	141.001	18
190	200	55	56.1662	61.8889	144	141.001	19
200	210	55	56.1662	61.8889	144	141.001	20
210	220	55	56.1662	61.8889	144	141.001	21
220	230	55	56.1662	61.8889	144	141.001	22
230	240	55	56.1662	61.8889	144	141.001	23

240	250	55	56.1662	61.8889	144	141.001	24
250	260	55	56.1662	61.8889	144	141.001	25
260	270	55	56.1662	61.8889	144	141.001	26
270	280	55	56.1662	61.8889	144	141.001	27
280	290	55	56.1662	61.8889	144	141.001	28
290	300	55	56.1662	61.8889	144	141.001	29
300	310	55	56.9533	63.6031	74	71.5095	30

Minimum	55	62
---------	----	----

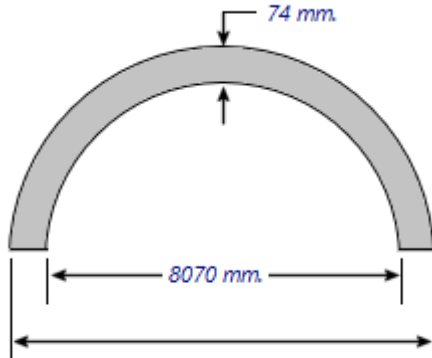
MAWP: 55 bars, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 1

Spherical Head Design Information:					
Design Pressure	P	55.001 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8070.000 mm.	Actual Stress		152.8 N./mm ²
Spec. Min. Thk t _e , or s _a		74.0000 mm.	Spec. Nominal Thk.		78.000 mm.
Surface Area		1062912.0 cm ²	SG of Contents		0.002
Element Volume		137615424.0 cm ³	Weight of Contents		247.5 Kg
Empty Weight		63046.1 Kg	Operating Weight		63293.7 Kg
			Comp. All. Ope. Stress		105.8 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 100.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 50.438 mm.

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 8074.0/2 \cdot (\exp(0.5 \cdot 55.001/(161.141 \cdot 1.0)) - 1) + 2.0 + 0.0$
 $= 69.4917 + 2.0 + 0.0 = 71.4917 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 161.1 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 8074.0)/8074.0) - 0.001$
 $= 56.969 - 0.001 = 56.968 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 174.997 \cdot 1.0 \cdot \ln((2 \cdot 74.0 + 8070.0)/8070.0)$
 $= 63.602 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 55.001 \cdot 8074.0^2 / (1.0 (8218.001^2 - 8074.0^2)) \\ &= 152.841 \text{ N./mm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 55.001 / (1.0 \cdot \ln((2 \cdot 72.0 + 8074.0) / 8074.0)) \\ &= 155.575 \text{ N./mm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.436 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.572 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 1062912 / 3.14159) \\ &= 11633.320 \text{ mm.} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

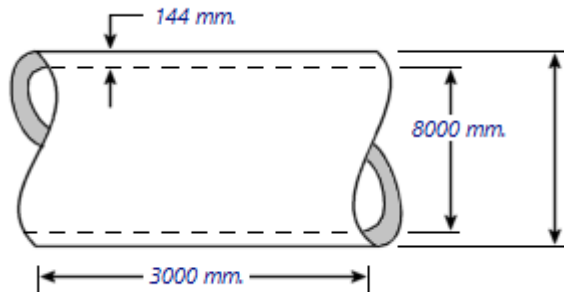
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 100.875 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 50.438 mm.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 2

Cylindrical Shell Design Information:					
Design Pressure	P	55.001 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.001/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0003 + 2.0 + 0.0 = 141.0003 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.001$$

$$= 56.168 - 0.001 = 56.1664 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.001 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0197 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.001 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7547 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

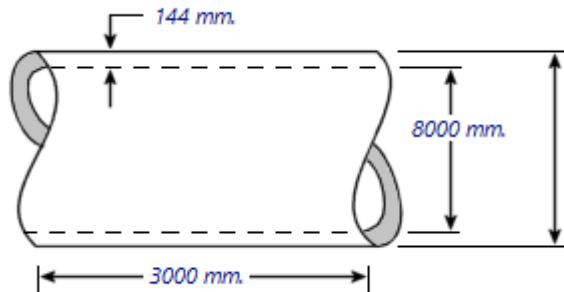
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 3

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

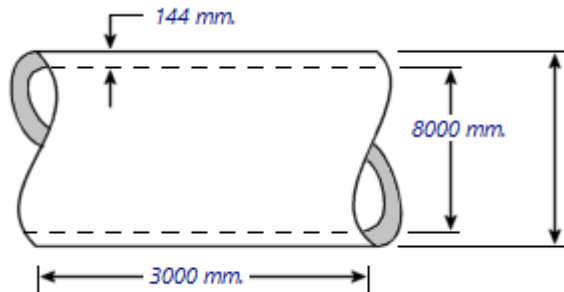
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 4

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

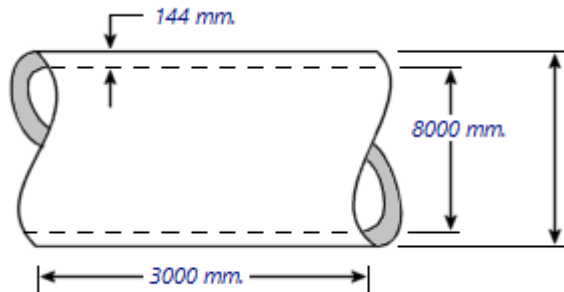
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 5

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

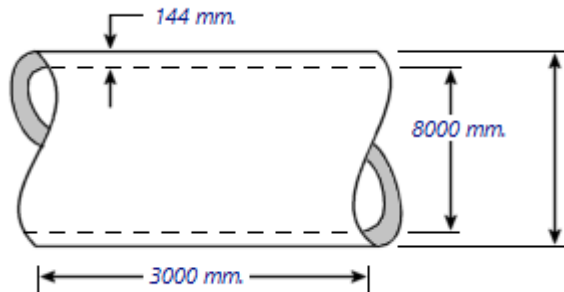
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 6

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

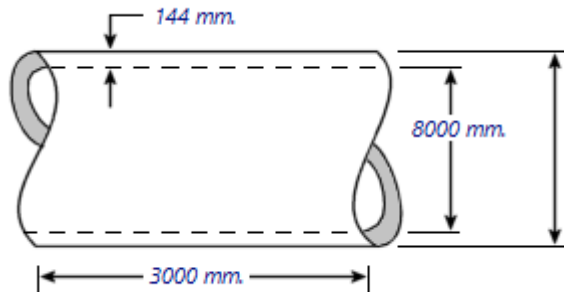
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 7

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

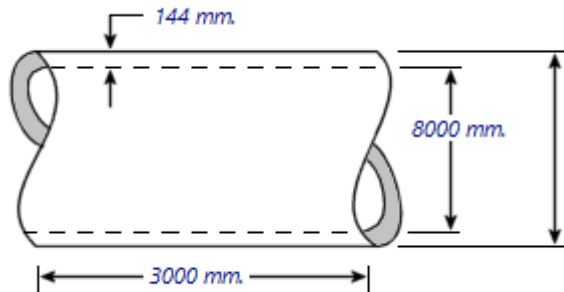
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 8

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

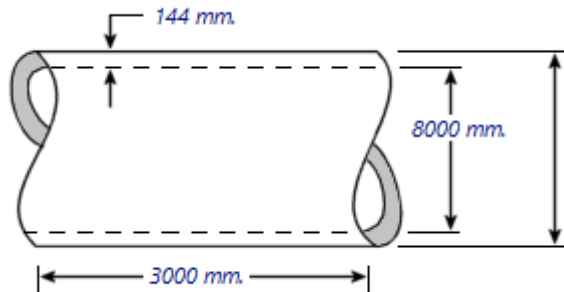
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 9

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

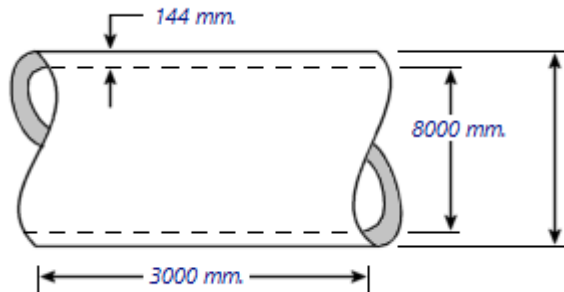
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 10

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

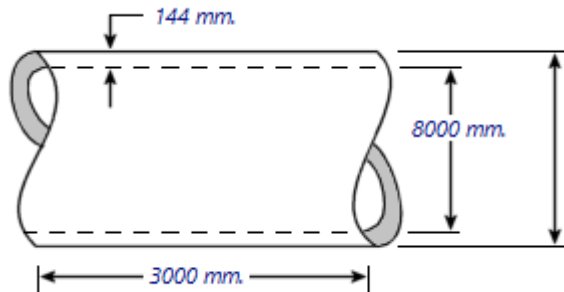
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 11

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

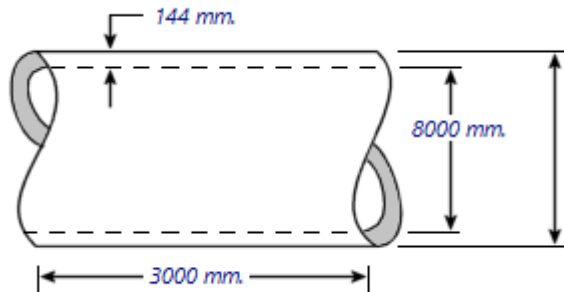
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 12

Cylindrical Shell Design Information:					
Design Pressure	P	55.000 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.000
Element Volume		150823456.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		85660.3 Kgf	Operating Weight		85660.3 Kgf
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.0/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 138.9964 + 2.0 + 0.0 = 140.9964 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.0$$

$$= 56.168 - 0.0 = 56.1678 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.0 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0158 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.0 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7506 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

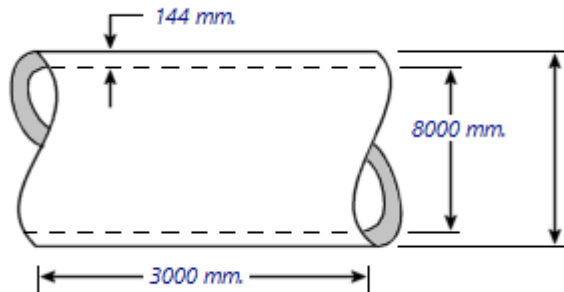
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 13

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

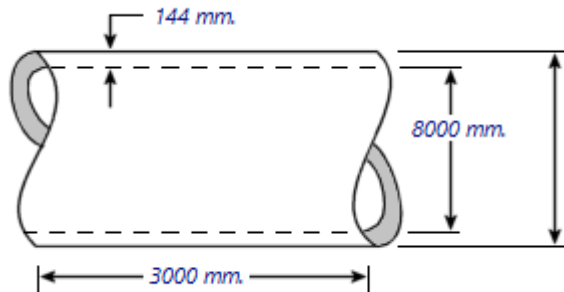
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 14

Cylindrical Shell Design Information:					
Design Pressure	P	55.000 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.000
Element Volume		150823456.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		85660.3 Kgf	Operating Weight		85660.3 Kgf
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:				
...		...		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.0/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 138.9964 + 2.0 + 0.0 = 140.9964 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.0$$

$$= 56.168 - 0.0 = 56.1678 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.0 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0158 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.0 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7506 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

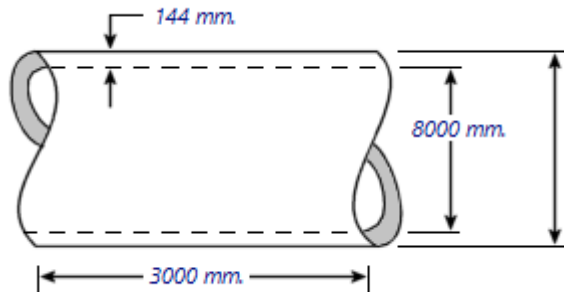
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 15

Cylindrical Shell Design Information:					
Design Pressure	P	55.000 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.000
Element Volume		150823456.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		85660.3 Kgf	Operating Weight		85660.3 Kgf
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.0/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 138.9964 + 2.0 + 0.0 = 140.9964 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.0$$

$$= 56.168 - 0.0 = 56.1678 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.0 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0158 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.0 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7506 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

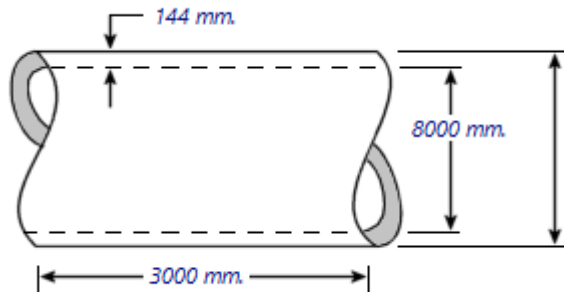
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 16

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

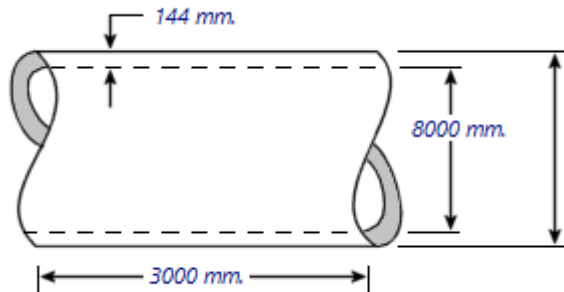
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 17

Cylindrical Shell Design Information:					
Design Pressure	P	55.000 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.000
Element Volume		150823456.0 cm ³	Weight of Contents		0.0 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85660.3 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.0/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 138.9964 + 2.0 + 0.0 = 140.9964 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.0$$

$$= 56.168 - 0.0 = 56.1678 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.0 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0158 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



= 55.0/(1.0 • Ln((2•142.0 + 8004.0)/8004.0))
= 157.7506 N./mm²

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate (50*tnom/Rf)*(1-Rf/Ro) 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

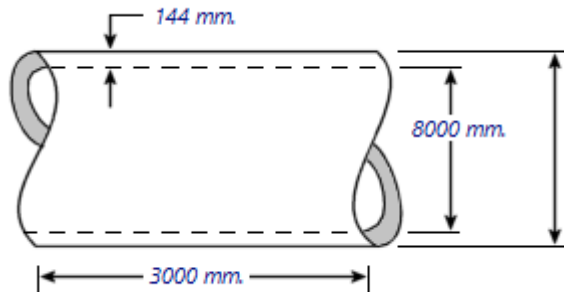
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was
specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 18

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

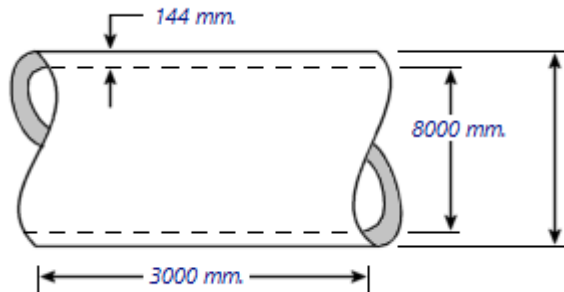
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 19

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

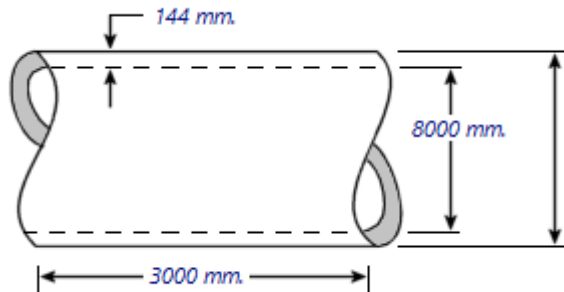
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 20

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

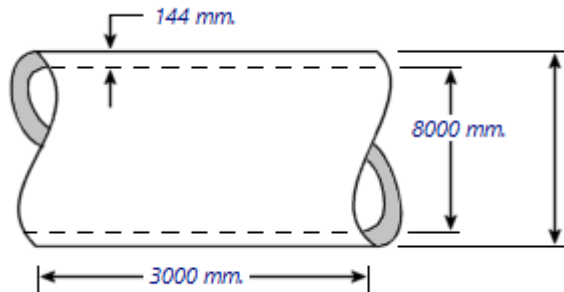
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 21

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

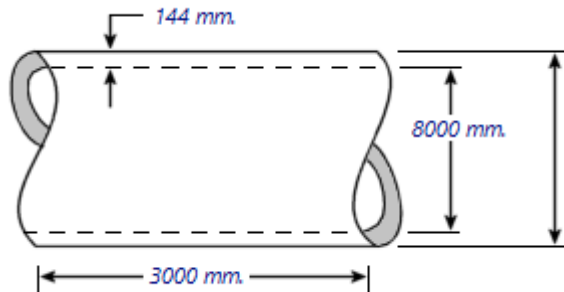
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 22

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

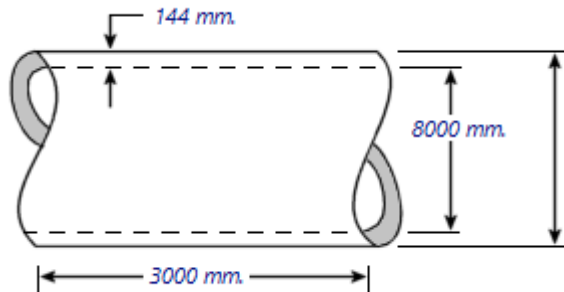
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 23

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or sa		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

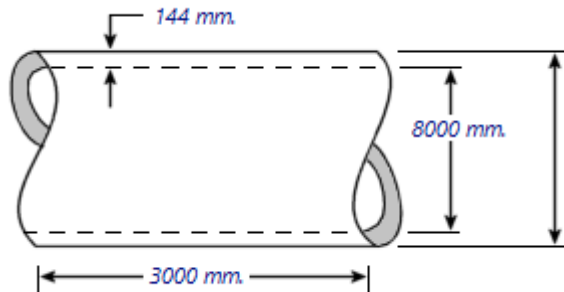
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 24

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

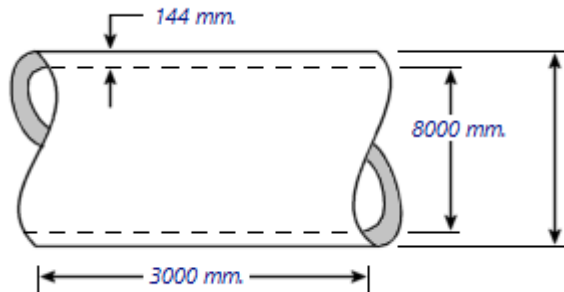
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 25

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg
Empty Weight		85660.3 Kg	Operating Weight		85931.6 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

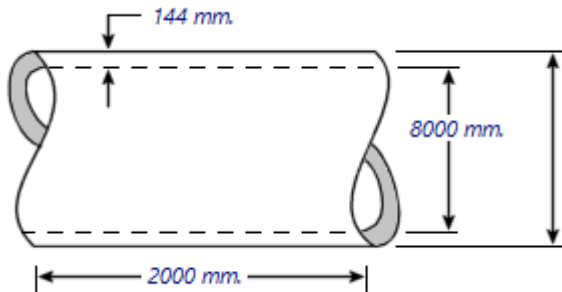
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 26

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		2000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		520750.4 cm ²	SG of Contents		0.002
Element Volume		100548976.0 cm ³	Weight of Contents		180.9 Kg _f
Empty Weight		57106.9 Kg _f	Operating Weight		57287.8 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

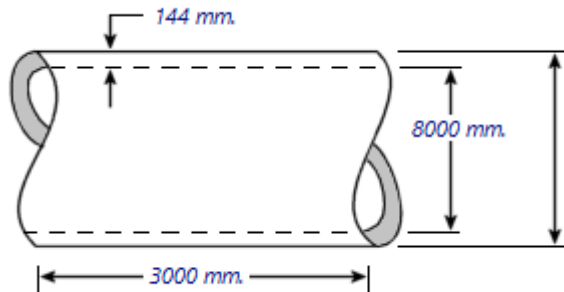
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 27

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

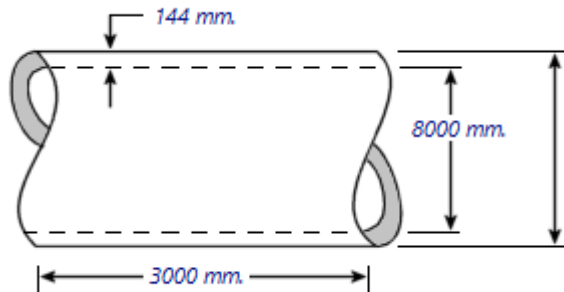
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 28

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

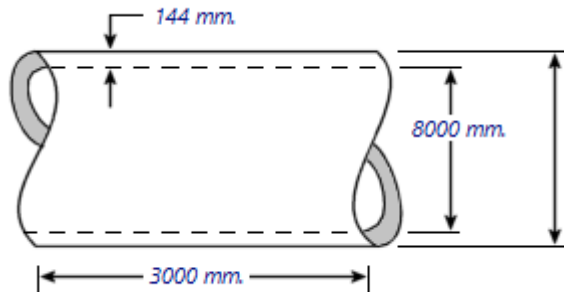
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 29

Cylindrical Shell Design Information:					
Design Pressure	P	55.002 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8000.000 mm.	Actual Stress		155.0 N./mm ²
Spec. Min. Thk t _e , or s _a		144.0000 mm.	Spec. Nominal Thk.		144.000 mm.
Surface Area		781125.6 cm ²	SG of Contents		0.002
Element Volume		150823456.0 cm ³	Weight of Contents		271.3 Kg _f
Empty Weight		85660.3 Kg _f	Operating Weight		85931.6 Kg _f
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 8004.0 \cdot (\exp(55.002/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 139.0007 + 2.0 + 0.0 = 141.0007 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 161.1 \cdot 1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0) - 0.002$$

$$= 56.168 - 0.002 = 56.1662 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 175.0 \cdot 1.0 \cdot \ln((2 \cdot 144.0 + 8000.0) / 8000.0)$$

$$= 61.8889 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 55.002 \cdot 8004.0 / (1.0 (8288.0 - 8004.0))$$

$$= 155.0202 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 55.002 / (1.0 \cdot \ln((2 \cdot 142.0 + 8004.0) / 8004.0))$$
$$= 157.7551 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.768 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

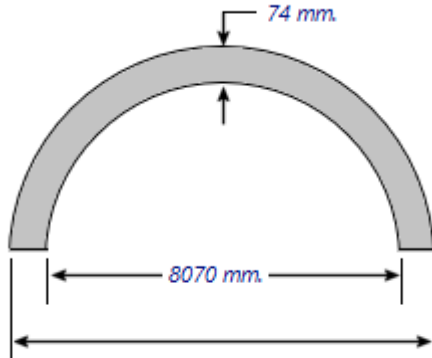
Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Element # 30

Spherical Head Design Information:					
Design Pressure	P	55.001 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		0.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	161.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	175.0 N./mm ²
Inside Diameter	Di	8070.000 mm.	Actual Stress		152.8 N./mm ²
Spec. Min. Thk t _e , or s _a		74.0000 mm.	Spec. Nominal Thk.		78.000 mm.
Surface Area		1062912.0 cm ²	SG of Contents		0.002
Element Volume		137615424.0 cm ³	Weight of Contents		247.5 Kg
Empty Weight		63042.9 Kg	Operating Weight		63290.4 Kg
			Comp. All. Ope. Stress		117.5 N./mm ²

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 100.875 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 50.438 mm.

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 8074.0/2 \cdot (\exp(0.5 \cdot 55.001/(161.1 \cdot 1.0)) - 1) + 2.0 + 0.0$
 $= 69.5095 + 2.0 + 0.0 = 71.5095 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 161.1 \cdot 1.0 \cdot \ln((2 \cdot 72.0 + 8074.0)/8074.0) - 0.001$
 $= 56.955 - 0.001 = 56.9533 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 175.0 \cdot 1.0 \cdot \ln((2 \cdot 74.0 + 8070.0)/8070.0)$
 $= 63.6031 \text{ bars}$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 55.001 \cdot 8074.0^2 / (1.0 (8218.001^2 - 8074.0^2)) \\ &= 152.841 \text{ N./mm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 55.001 / (1.0 \cdot \ln((2 \cdot 72.0 + 8074.0) / 8074.0)) \\ &= 155.575 \text{ N./mm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	1.436 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	36.572 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 1062912 / 3.14159) \\ &= 11633.320 \text{ mm.} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 100.875 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 50.438 mm.

Material treatment options: +IT

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 55.2741(1.077)$
 $= 74.442 \text{ bars}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From	To	Stress	Allowable	Ratio	Pressure
10	20	212.8	247.0	0.862	75.23
20	30	215.8	247.0	0.874	75.23
30	40	215.8	247.0	0.874	75.23
40	50	215.8	247.0	0.874	75.23
50	60	215.8	247.0	0.874	75.23
60	70	215.8	247.0	0.874	75.23
70	80	215.8	247.0	0.874	75.23
80	90	215.8	247.0	0.874	75.23
90	100	215.8	247.0	0.874	75.23
100	110	215.8	247.0	0.874	75.23
110	120	215.8	247.0	0.874	75.23
120	130	215.8	247.0	0.874	75.23
130	140	215.8	247.0	0.874	75.23
140	150	215.8	247.0	0.874	75.23
150	160	215.8	247.0	0.874	75.23
160	170	215.8	247.0	0.874	75.23
170	180	215.8	247.0	0.874	75.23
180	190	215.8	247.0	0.874	75.23
190	200	215.8	247.0	0.874	75.23
200	210	215.8	247.0	0.874	75.23
210	220	215.8	247.0	0.874	75.23
220	230	215.8	247.0	0.874	75.23
230	240	215.8	247.0	0.874	75.23
240	250	215.8	247.0	0.874	75.23
250	260	215.8	247.0	0.874	75.23
260	270	215.8	247.0	0.874	75.23
270	280	215.8	247.0	0.874	75.23
280	290	215.8	247.0	0.874	75.23
290	300	215.8	247.0	0.874	75.23
300	310	212.8	247.0	0.862	75.23

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1	Nozzle	165.00	153.14	1.077
N2	Nozzle	165.00	153.14	1.077
N1	Nozzle	165.00	153.14	1.077
M2	Nozzle	165.00	153.14	1.077
K2	Nozzle	165.00	153.14	1.077
K1	Nozzle	165.00	153.14	1.077
M3	Nozzle	165.00	153.14	1.077
N3	Nozzle	165.00	153.14	1.077

Minimum

1.077

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
	175.00	161.14	1.086

[illegible]

Minimum

1.086

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M1	181.86	237.50	0.766
N2	205.13	237.50	0.864
N1	196.33	237.50	0.827
M2	181.86	237.50	0.766
K2	17.23	237.50	0.073
K1	17.23	237.50	0.073
M3	181.86	237.50	0.766
N3	20.29	237.50	0.085

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	8218	72	No Calc	No Calc
20	30	85690	8288	142	No Calc	No Calc
30	40	85690	8288	142	No Calc	No Calc
40	50	85690	8288	142	No Calc	No Calc
50	60	85690	8288	142	No Calc	No Calc
60	70	85690	8288	142	No Calc	No Calc
70	80	85690	8288	142	No Calc	No Calc
80	90	85690	8288	142	No Calc	No Calc
90	100	85690	8288	142	No Calc	No Calc
100	110	85690	8288	142	No Calc	No Calc
110	120	85690	8288	142	No Calc	No Calc
120	130	85690	8288	142	No Calc	No Calc
130	140	85690	8288	142	No Calc	No Calc
140	150	85690	8288	142	No Calc	No Calc
150	160	85690	8288	142	No Calc	No Calc
160	170	85690	8288	142	No Calc	No Calc
170	180	85690	8288	142	No Calc	No Calc
180	190	85690	8288	142	No Calc	No Calc
190	200	85690	8288	142	No Calc	No Calc
200	210	85690	8288	142	No Calc	No Calc
210	220	85690	8288	142	No Calc	No Calc
220	230	85690	8288	142	No Calc	No Calc
230	240	85690	8288	142	No Calc	No Calc
240	250	85690	8288	142	No Calc	No Calc
250	260	85690	8288	142	No Calc	No Calc
260	270	85690	8288	142	No Calc	No Calc
270	280	85690	8288	142	No Calc	No Calc
280	290	85690	8288	142	No Calc	No Calc
290	300	85690	8288	142	No Calc	No Calc
300	310	No Calc	8218	72	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	74	No Calc	...	27.0658
20	30	144	No Calc	...	8.73431
30	40	144	No Calc	...	8.79424
40	50	144	No Calc	...	8.79424
50	60	144	No Calc	...	8.79424
60	70	144	No Calc	...	8.79424
70	80	144	No Calc	...	8.79424
80	90	144	No Calc	...	8.79424
90	100	144	No Calc	...	8.79424
100	110	144	No Calc	...	8.79424
110	120	144	No Calc	...	8.79424
120	130	144	No Calc	...	8.79424
130	140	144	No Calc	...	8.79424
140	150	144	No Calc	...	8.79424
150	160	144	No Calc	...	8.79424
160	170	144	No Calc	...	8.79424
170	180	144	No Calc	...	8.79424
180	190	144	No Calc	...	8.79424
190	200	144	No Calc	...	8.79424
200	210	144	No Calc	...	8.79424
210	220	144	No Calc	...	8.79424



220	230	144	No Calc	...	8.79424
230	240	144	No Calc	...	8.79424
240	250	144	No Calc	...	8.79424
250	260	144	No Calc	...	8.79424
260	270	144	No Calc	...	8.79424
270	280	144	No Calc	...	8.79424
280	290	144	No Calc	...	8.79424
290	300	144	No Calc	...	8.79424
300	310	74	No Calc	...	27.0665

Minimum

9

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	85690	No Calc	No Calc	No Calc
30	40	85690	No Calc	No Calc	No Calc
40	50	85690	No Calc	No Calc	No Calc
50	60	85690	No Calc	No Calc	No Calc
60	70	85690	No Calc	No Calc	No Calc
70	80	85690	No Calc	No Calc	No Calc
80	90	85690	No Calc	No Calc	No Calc
90	100	85690	No Calc	No Calc	No Calc
100	110	85690	No Calc	No Calc	No Calc
110	120	85690	No Calc	No Calc	No Calc
120	130	85690	No Calc	No Calc	No Calc
130	140	85690	No Calc	No Calc	No Calc
140	150	85690	No Calc	No Calc	No Calc
150	160	85690	No Calc	No Calc	No Calc
160	170	85690	No Calc	No Calc	No Calc
170	180	85690	No Calc	No Calc	No Calc
180	190	85690	No Calc	No Calc	No Calc
190	200	85690	No Calc	No Calc	No Calc
200	210	85690	No Calc	No Calc	No Calc
210	220	85690	No Calc	No Calc	No Calc
220	230	85690	No Calc	No Calc	No Calc
230	240	85690	No Calc	No Calc	No Calc
240	250	85690	No Calc	No Calc	No Calc
250	260	85690	No Calc	No Calc	No Calc
260	270	85690	No Calc	No Calc	No Calc
270	280	85690	No Calc	No Calc	No Calc
280	290	85690	No Calc	No Calc	No Calc
290	300	85690	No Calc	No Calc	No Calc
300	310	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	63046.1	137615424	61459.7	137820144	...
20	30	85660.3	150823456	84491.4	150974320	...
30	40	85660.3	150823456	84491.4	150974320	...
40	50	85660.3	150823456	84491.4	150974320	...
50	60	85660.3	150823456	84491.4	150974320	...
60	70	85660.3	150823456	84491.4	150974320	...
70	80	85660.3	150823456	84491.4	150974320	...
80	90	85660.3	150823456	84491.4	150974320	...
90	100	85660.3	150823456	84491.4	150974320	...
100	110	85660.3	150823456	84491.4	150974320	...
110	120	85660.3	150823456	84491.4	150974320	...
120	130	85660.3	150823456	84491.4	150974320	...
130	140	85660.3	150823456	84491.4	150974320	...
140	150	85660.3	150823456	84491.4	150974320	...
150	160	85660.3	150823456	84491.4	150974320	...
160	170	85660.3	150823456	84491.4	150974320	...
170	180	85660.3	150823456	84491.4	150974320	...
180	190	85660.3	150823456	84491.4	150974320	...
190	200	85660.3	150823456	84491.4	150974320	...
200	210	85660.3	150823456	84491.4	150974320	...
210	220	85660.3	150823456	84491.4	150974320	...
220	230	85660.3	150823456	84491.4	150974320	...
230	240	85660.3	150823456	84491.4	150974320	...
240	250	85660.3	150823456	84491.4	150974320	...
250	260	85660.3	150823456	84491.4	150974320	...
260	270	57106.9	100548976	56327.6	100649552	...
270	280	85660.3	150823456	84491.4	150974320	...
280	290	85660.3	150823456	84491.4	150974320	...
290	300	85660.3	150823456	84491.4	150974320	...
300	310	63042.9	137615424	61456.5	137820144	...
Total		2496024	0.445E+10	2460511	0.445E+10	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	247.54	-1345	...	...	Liquid: 10
10	Wght	7830	-4035	...	...	cut back-1
20	Plat	10688.2	5000	4220.2	...	PLAT:[1 OF 1]
20	Liqd	271.298	1500	...	...	Liquid: 20
20	Wght	100	3000	4500	...	MH Davit
20	Wght	650	3000	4209	...	PL clip
30	Liqd	271.298	1500	-440	...	Liquid: 30
30	Nozl	1604.47	1500	4304.8	...	M1
30	Wght	500	1500	...	...	INT Ladder-1
40	Sadl	14029.4	2000	4353	...	Saddle-1
40	Liqd	271.298	1500	-440	...	Liquid: 40
40	Nozl	3255.88	2499.98	4533.4	...	N2
40	Nozl	962.895	499.975	4254	...	N1
40	Wght	7000	2000	-4000	...	FP-1
40	Wght	2000	3000	...	...	Fire Cleat-1
40	Wght	3500	...	4500	...	CV-1
40	Wght	500	500	4500	...	Valve
50	Liqd	271.298	1500	-440	...	Liquid: 50
60	Liqd	271.298	1500	-440	...	Liquid: 60
60	Wght	5000	-0.024418	...	...	misc-1
60	Wght	1700	-0.024418	...	...	Fire cleat-1
70	Liqd	271.298	1500	-440	...	Liquid: 70
80	Liqd	271.298	1500	-440	...	Liquid: 80
90	Liqd	271.298	1500	-440	...	Liquid: 100



100	Liqd	271.298	1500	-440	...	Liquid: 110
110	Liqd	271.298	1500	-440	...	Liquid: 120
120	Sadl	13606.8	333	4297	...	New Saddle
120	Wght	7000	333	-4000	...	FP2
130	Liqd	271.298	1500	-440	...	Liquid: 130
160	Plat	2013.79	1500	4220.2	...	PLAT:[1 OF 1]
160	Liqd	271.298	1500	-440	...	Liquid: 140
160	Nozl	1604.47	1500	4304.8	...	M2
160	Wght	250	...	4200	...	PL clip-3
160	Wght	100	1500	4500	...	MH Devit
160	Wght	500	1500	...	...	INT Ladder-2
180	Liqd	271.298	1500	-440	...	Liquid: 160
190	Sadl	13184.2	1666	4241	...	New Saddle
190	Liqd	271.298	1500	-440	...	Liquid: 170
190	Wght	7000	1666	-4000	...	FP3
200	Liqd	271.298	1500	-440	...	Liquid: 180
210	Liqd	271.298	1500	-440	...	Liquid: 190
220	Liqd	271.298	1500	-440	...	Liquid: 210
230	Liqd	271.298	1500	-440	...	Liquid: 220
240	Liqd	271.298	1500	-440	...	Liquid: 230
240	Wght	5000	2999.96	...	...	misc 2
240	Wght	1700	2999.96	...	...	Fire cleat-2
250	Liqd	271.298	1500	-440	...	Liquid: 240
260	Liqd	180.866	1000	-440	...	Liquid: 250
270	Sadl	12765.3	1000	4185.5	...	New Saddle
270	Plat	10688.2	5000	4220.2	...	PLAT:[1 OF 1]
270	Liqd	271.298	1500	-440	...	Liquid: 260
270	Nozl	33.2639	500	4025.4	...	K2
270	Nozl	33.2639	2500	4025.4	...	K1
270	Wght	650	2999.96	4209	...	PL clip-2
270	Wght	7000	1000	-4000	...	FP4
270	Wght	2000	...	...	...	Fire cleat-2
280	Liqd	271.298	1500	-440	...	Liquid: 270
280	Nozl	1604.47	1500	4304.8	...	M3
280	Wght	50	1999.96	4500	...	MH davit
280	Wght	500	1500	...	...	INT Ladder-3
290	Liqd	271.298	1500	-440	...	Liquid: 280
290	Nozl	40.5593	1500	-4038.1	...	N3
300	Liqd	247.54	1345	...	...	Liquid: 290
300	Wght	7830	4035	...	...	cut back 2

Total Weight of Each Detail Type:

Saddles	53585.8
Platforms	23390.2
Liquid	6915.8
Nozzles	9139.3
Weights	68360.0

Sum of the Detail Weights 161391.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	2496024.5	2496024.5	2496024.5	2496024.5	2496024.5	2496024.5
Saddles	53585.8	53585.8	53585.8	53585.8	53585.8	53585.8
Nozzles	9139.3	9139.3	9139.3	9139.3	9139.3	9139.3
Wld Weights	35860.0	35860.0	35860.0	35860.0	35860.0	35860.0
Platforms	...	...	23390.2	23390.2	23390.2	23390.2
Empty Weights	...	...	...	32500.0	32500.0	...
Test Weights	...	32500.0	...	...	...	...
Ope Weights	...	...	...	...	...	32500.0
Ope. Liquid	...	...	...	...	...	6915.8
Test Liquid	...	4445297.0	...	...	...	...
Totals	2594609.8	7072406.5	2617999.8	2650499.8	2650499.8	2657415.5

Field Installation Options:



Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2594609.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	7072406.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	2617999.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	2650499.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	2650499.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	2657415.5 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	2621902.2 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	1062912	1022979
20	30	781126	753982
30	40	781126	753982
40	50	781126	753982
50	60	781126	753982
60	70	781126	753982
70	80	781126	753982
80	90	781126	753982
90	100	781126	753982
100	110	781126	753982
110	120	781126	753982
120	130	781126	753982
130	140	781126	753982
140	150	781126	753982
150	160	781126	753982
160	170	781126	753982
170	180	781126	753982
180	190	781126	753982
190	200	781126	753982
200	210	781126	753982
210	220	781126	753982
220	230	781126	753982
230	240	781126	753982
240	250	781126	753982
250	260	781126	753982
260	270	520750	502655
270	280	781126	753982
280	290	781126	753982
290	300	781126	753982
300	310	1062912	1022979

Total	23736970.0	22906138.0 cm ²
Total	25550.3	24656.0 ft ²

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
M1	95.26	102.00	85	600	GR 1.1	...	...	...	...
N2	95.26	102.00	85	600	GR 1.1	...	...	...	...
N1	95.26	102.00	85	600	GR 1.1	...	...	...	...
M2	95.26	102.00	85	600	GR 1.1	...	...	...	...
K2	95.26	102.00	85	600	GR 1.1	...	...	...	...
K1	95.26	102.00	85	600	GR 1.1	...	...	...	...
M3	95.26	102.00	85	600	GR 1.1	...	...	...	...
Min Rating	95.261	102.000, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	500 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.954) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.902)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.620
Structure Height to Diameter ratio	10.621

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Z_g = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 4880.0 + 500.0 = 5380.0$ mm.
 $= 17.651$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (17.651 ft.) > 15 ft.
 $= 2.41 (z / Z_g)^{2/\alpha}$
 $= 2.41 (17.651 / 1935.0)^{2/11.5}$
 $= 1.065$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:
 $K_1 = 0$, $K_2 = 0$, $K_3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.065 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 34.1 \text{ psf } [166.496 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 34.1 \cdot 0.85 \cdot 0.62 \cdot 285.474$$

$$= 5131.2 \text{ lbs. } [2327.5 \text{ Kgf}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Node 10 to 20	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 20 to 30	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 30 to 40	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 40 to 50	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 50 to 60	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 60 to 70	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 70 to 80	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 80 to 90	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 90 to 100	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 100 to 110	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 110 to 120	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 120 to 130	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 130 to 140	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 140 to 150	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 150 to 160	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 160 to 170	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 170 to 180	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 180 to 190	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 190 to 200	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 200 to 210	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 210 to 220	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 220 to 230	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 230 to 240	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 240 to 250	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 250 to 260	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 260 to 270	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 270 to 280	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 280 to 290	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 290 to 300	5380.0	0.000	0.000	0.000	1.065	1.000	166.496
Node 300 to 310	5380.0	0.000	0.000	0.000	1.065	1.000	166.496

Platform Load Calculations

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf	Cf
PLAT:[1 OF 1]	297960.00	5380.00	166.50	4960.91	2.00
PLAT:[1 OF 1]	129960.00	5380.00	166.50	2163.78	2.00
PLAT:[1 OF 1]	297960.00	5380.00	166.50	4960.91	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
cut back-1	0.00	5380.00	166.50	0.00
MH Davit	0.00	9880.00	183.80	0.00
PL clip	0.00	9589.00	182.68	0.00
INT Ladder-1	0.00	5380.00	166.50	0.00



FP-1	0.00	500.00	166.50	0.00
Fire Cleat-1	0.00	5380.00	166.50	0.00
CV-1	0.00	9880.00	183.80	0.00
Valve	0.00	9880.00	183.80	0.00
misc-1	0.00	5380.00	166.50	0.00
Fire cleat-1	0.00	5380.00	166.50	0.00
FP2	0.00	500.00	166.50	0.00
PL clip-3	0.00	9580.00	182.65	0.00
MH Devit	0.00	9880.00	183.80	0.00
INT Ladder-2	0.00	5380.00	166.50	0.00
FP3	0.00	500.00	166.50	0.00
misc 2	0.00	5380.00	166.50	0.00
Fire cleat-2	0.00	5380.00	166.50	0.00
PL clip-2	0.00	9589.00	182.68	0.00
FP4	0.00	500.00	166.50	0.00
Fire cleat-2	0.00	5380.00	166.50	0.00
MH davit	0.00	9880.00	183.80	0.00
INT Ladder-3	0.00	5380.00	166.50	0.00
cut back 2	0.00	5380.00	166.50	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	5380	9861.6	265214	166.496	2560.27
20	30	5380	9945.6	298368	166.496	8337.33
30	40	5380	9945.6	298368	166.496	2880.32
40	50	5380	9945.6	298368	166.496	2880.32
50	60	5380	9945.6	298368	166.496	2880.32
60	70	5380	9945.6	298368	166.496	2880.32
70	80	5380	9945.6	298368	166.496	2880.32
80	90	5380	9945.6	298368	166.496	2880.32
90	100	5380	9945.6	298368	166.496	2880.32
100	110	5380	9945.6	298368	166.496	2880.32
110	120	5380	9945.6	298368	166.496	2880.32
120	130	5380	9945.6	298368	166.496	2880.32
130	140	5380	9945.6	298368	166.496	2880.32
140	150	5380	9945.6	298368	166.496	2880.32
150	160	5380	9945.6	298368	166.496	2880.32
160	170	5380	9945.6	298368	166.496	5260.48
170	180	5380	9945.6	298368	166.496	2880.32
180	190	5380	9945.6	298368	166.496	2880.32
190	200	5380	9945.6	298368	166.496	2880.32
200	210	5380	9945.6	298368	166.496	2880.32
210	220	5380	9945.6	298368	166.496	2880.32
220	230	5380	9945.6	298368	166.496	2880.32
230	240	5380	9945.6	298368	166.496	2880.32
240	250	5380	9945.6	298368	166.496	2880.32
250	260	5380	9945.6	298368	166.496	2880.32
260	270	5380	9945.6	198912	166.496	1920.22
270	280	5380	9945.6	298368	166.496	8337.33
280	290	5380	9945.6	298368	166.496	2880.32
290	300	5380	9945.6	298368	166.496	2880.32
300	310	5380	9861.6	265214	166.496	2560.27

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	2.500
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(28.14960.75)$
 = 0.244 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.244$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 2568316 \cdot 1.5$
 = 251181.328 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 2568316$
 = 111636.1 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	4035	75538.7	7448.5
20	30	4000	75538.7	7383.89
30	40	4000	75538.7	7383.89
40	50	4000	75538.7	7383.89
40	50	4000	75538.7	7383.89
50	60	4000	75538.7	7383.89



60	70	4000	75538.7	7383.89
70	80	4000	75538.7	7383.89
80	90	4000	75538.7	7383.89
90	100	4000	75538.7	7383.89
100	110	4000	75538.7	7383.89
110	120	4000	75538.7	7383.89
120	0	4000	75538.7	7383.89
120	130	4000	75538.7	7383.89
130	140	4000	75538.7	7383.89
140	150	4000	75538.7	7383.89
150	160	4000	75538.7	7383.89
160	170	4000	75538.7	7383.89
170	180	4000	75538.7	7383.89
180	190	4000	75538.7	7383.89
190	200	4000	75538.7	7383.89
190	200	4000	75538.7	7383.89
200	210	4000	75538.7	7383.89
210	220	4000	75538.7	7383.89
220	230	4000	75538.7	7383.89
230	240	4000	75538.7	7383.89
240	250	4000	75538.7	7383.89
250	260	4000	75538.7	7383.89
260	270	4000	75538.7	7383.89
270	0	4000	75538.7	7383.89
270	280	4000	75538.7	7383.89
280	290	4000	75538.7	7383.89
290	300	4000	75538.7	7383.89
300	310	4035	75538.7	7448.5

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	40621.406 mm.
Center of Gravity of the Platforms	42129.133 mm.
Center of Gravity of the Liquid	41892.289 mm.
Center of Gravity of the Nozzles	26832.273 mm.
Center of Gravity of the Added Weights (Operating)	39192.238 mm.
Center of Gravity of the Added Weights (Empty)	39192.238 mm.
Center of Gravity of Bare Shell New and Cold	41499.934 mm.
Center of Gravity of Bare Shell Corroded	41499.941 mm.
Vessel CG in the Operating Condition	41377.332 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	41377.629 mm.
Vessel CG in the Test Condition	41454.277 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (900.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

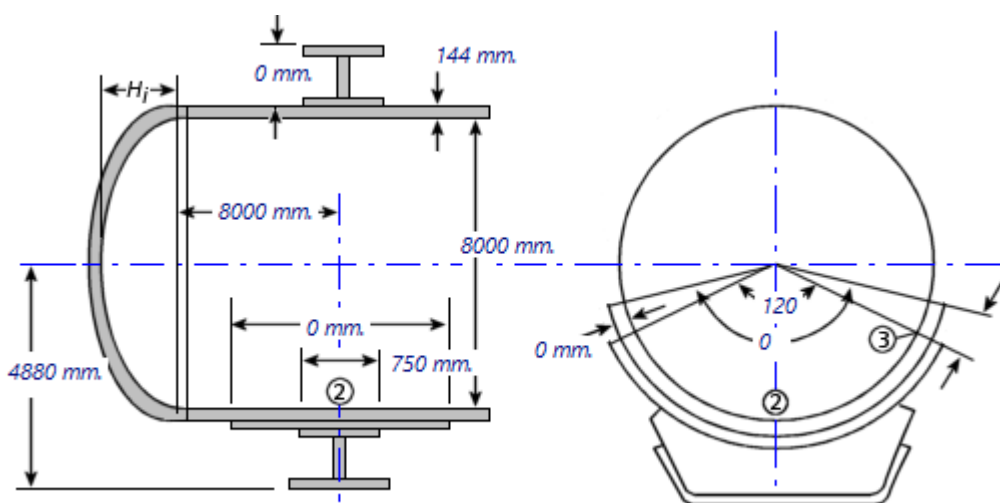
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(750.0 + 1.56 \cdot \sqrt{4073.0002 \cdot 142.0}, 2 \cdot 8000.0)$$

$$= 1936.3859 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	4073.00	mm.
Shell Thickness used in this Case	t	144.000	mm.
Stiffened Vessel Length per 4.15.6	L	82999.98	mm.
Distance from Saddle to Vessel tangent	a1 or a	8000.00	mm.
Saddle Width	b1 or b	750.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4073.00	mm.
Shell Allowable Stress used in Calculation		161.10	N./mm ²
Head Allowable Stress used in Calculation		161.14	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4880.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	797149.56	Kgf
Pressure used in Analysis	P	55.001 bars

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	66.76	161.10



Long. Stress at Bottom of Midspan	91.01	161.10
Long. Stress at Top of Saddles	90.92	161.10
Long. Stress at Bottom of Saddles	72.21	161.10
Tangential Shear in Shell	11.99	128.88
Circ. Stress at Horn of Saddle	37.84	201.38
Circ. Compressive Stress in Shell	2.16	161.10

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(98103.7/4 + 0) \cdot 4880.0/7229.5811$
 $= 49665.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(16865.43, 0) \cdot 4880.0/22333.0039$
 $= 3685.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,\text{eq}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(251181.36, 0) \cdot 4880.0/22333.0039$
 $= 54885.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(251181/4 + 0) \cdot 4880.0/7229.5811$
 $= 127161.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 669988 + \max(3685, 49665, 54886, 127161)$
 $= 797149.6 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1632.731(14.4 + \pi/4(8.288 \cdot 1.2)^2)$
 $= 165390.922 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	811179.06	Kgf
Transverse Shear Load	62795.34	Kgf
Longitudinal Shear Load	251181.36	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(8000.0/2 + 144.0 + 30.0)\sin(120.0/2), 2 \cdot 4073.0)$
 $= 7229.581 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -797150 \cdot 8000.0 [1 - (1 - 8000.0/82999.98 + (4073.0^2 - 4073.0^2)/(2 \cdot 8000.0 \cdot 82999.98))/(1 + (4 \cdot 4073.0)/(3 \cdot 82999.98))]$
 $= -968574.2 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/(3L)) - 4a/L)$
 $= 797150 \cdot 83000/4((1 + 2(4073^2 - 4073^2)/(83000^2))/(1 + (4 \cdot 4073)/(3 \cdot 83000)) - 4 \cdot 8000/83000)$



= 9148036.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 55.001 \cdot 4073.0 / (2 \cdot 142.0) - 9148036 / (\pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 66.76 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 55.001 \cdot 4073.0 / (2 \cdot 142.0) + 9148036 / (\pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 91.01 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 55.001 \cdot 4073.0 / (2 \cdot 142.0) - 968574.2 / (0.1066 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 90.92 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 55.001 \cdot 4073.0 / (2 \cdot 142.0) + 968574.2 / (0.1923 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 72.21 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 797150(82999.98 - 2 \cdot 8000.0) / (82999.98 + (4 \cdot 4073.0/3))$$

$$= 603964.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 603965 / (4073.0002 \cdot 142.0)$$

$$= 11.99 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(4073.0 \cdot 142.0)}$$

$$= 593.193 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 797150 \cdot 0.1 / (142.0(750.0 + 593.19 + 593.19))$$

$$= -2.16 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -797150 / (4 \cdot 142.0(750.0 + 593.193 + 593.193)) - 3 \cdot 0.0529 \cdot 797150 / (2 \cdot 142.0^2)$$

$$= -37.84 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(750.0 + 1.56 \cdot \sqrt{(4073.0 \cdot 142.0)}, 2 \cdot 8000.0)$$

$$= 1936.39 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7229.9995	mm.
Baseplate Thickness	Bpthk	130.0000	mm.
Baseplate Width	Bpwid	750.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	12	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	576.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$



$$= ((1.56 \cdot \sqrt{(4002.0 \cdot 142.0)} + 900.0) 142.0^3) / 12$$

$$= 49534.836 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 49534.836 + 2947.92 (257.427 - 71.0)^2$$

$$= 1074087.500 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	2076.0	142.0	71.0	2947.9	20930232.0	0.107E+07
Wearplate	900.0	30.0	157.0	270.0	4239000.0	0.274E+05
Web	30.0	576.0	460.0	172.8	7948807.0	0.119E+06
BasePlate	750.0	130.0	813.0	975.0	79267520.0	0.302E+07
Totals	...	...	...	4365.7	112385552.0	0.424E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 0.11239E+09 / 4365.72$$

$$= 257.427 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 797150$$

$$= 162237.2188 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 11.2218 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 4880.0 - 4002.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 1570.373 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 254777.9375 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 15.1574 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(797150 + 14029)750.0 / (4 \cdot 7230.0 \cdot 174.663))}$$

$$= 59.527 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (4000.0 + 144.0 + 30.0)$$

$$= 7229.580 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 7229.5801 / (12 - 1)$$

$$= 657.235 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 657.2346 \cdot 750.0 / 2$$

$$= 2464.630 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 811179 / (7230.0 \cdot 750.0)$$

$$= 14.960 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 2464.6 \cdot 14.96$$

$$= 36869.637 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 216.0 + 98.585$$

$$= 314.585 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 36869.6 / 314.5852$$

$$= 11.494 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	750.0	...	225.0	...	0.105E+06

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 750.0 - 30.0$$

$$= 720.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 314.585$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 750.0 / 2$$

$$= 375.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(105468.6 / 314.585)}$$

$$= 183.102 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 4880.0 - \cos(120.0 / 2) (4000.0 + 144.0 + 30.0) - 130.0$$

$$= 2663.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (4000.0 + 144.0 + 30.0)$$

$$= 7229.580 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 7230.0 - 50.8$$

$$= 7179.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (7229.58 - 7179.2) / 2$$

$$= 25.190 \text{ mm.}$$



Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(2663.0^2 + 25.19^2)}$$

$$= 2663.119 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20219\text{E}+09 / 262.0)}$$

$$= 123.428$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2663.119 / 183.102$$

$$= 14.544$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 251181.4 / (2 \cdot 7230.0) \cdot 657.235 \cdot 2663.12 / 2$$

$$= 15202.281 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($14.5445 < 123.4277$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.54)^2 / (2 \cdot 123.43^2)) 262 /$$

$$(5/3 + 3 \cdot (14.54) / (8 \cdot 123.43) - (14.54^3) / (8 \cdot 123.43^3))$$

$$= 152.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.49 / 152.09 + (15202.28 \cdot 375.0 / 1.0547\text{E}+10) / 174.66$$

$$= 0.379$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	720.0	0.0	216.0	0.0	0.105E+06
Web	657.2	30.0	0.0	197.2	0.0	148.
Totals	...	...	...	413.2	...	0.106E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 413.17$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 750.0 / 2$$

$$= 375.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 4880.0 - \sqrt{(4174.0 + 30.0)^2 - (657.235/2)^2} - 130.0$$

$$= 588.956 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(105609.8 / 413.17)}$$

$$= 159.878 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 588.956 / 159.878$$

$$= 3.684$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 251181.359 / (2 \cdot 7230.0)$$

$$= 17.371 \text{ Kg/mm.}$$



Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 17.371 \cdot 657.235 \cdot 588.956$$

$$= 6724.048 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 6724.048 / 2816263$$

$$= 23.414 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.6838 < 123.4277$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.68)^2 / (2 \cdot 123.43^2)) 262 / (5/3 + 3 \cdot (3.68) / (8 \cdot 123.43) - (3.68^3) / (8 \cdot 123.43^3))$$

$$= 156.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 17.83 / 156.08 + (6724.05 \cdot 375.0 / 105609.773) / 174.66$$

$$= 0.248$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	12
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	193.87 N./mm ²
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	17.2000 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	17.2000 cm ²
Saddle Load QO (Weight)	QO	684017.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	54885.8 Kg
Maximum Transverse Force	Ft	62795.3 Kg
Maximum Longitudinal Force	Fl	251181.4 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 68275.4 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 251181.4 / (17.2 \cdot 12)$$

$$= 119.3 \text{ N./mm}^2. \text{ Must be less than } 129.3 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 62795.3 / (17.2 \cdot 12)$$

$$= 29.8 \text{ N./mm}^2. \text{ Must be less than } 129.3 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 251181.36 / (129.27 \cdot 12.0)$$

$$= 15.8795 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 4880.0 \cdot 62795.34 + 0.0$$

$$= 306447.31 \text{ Kg-m.}$$

Eccentricity (e):



= Rmom / QO
 = 306447.31/684018
 = 448.00 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 15.8795, 0.0]
 = 15.8795 cm²

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (900.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(750.0 + 1.56 \cdot \sqrt{4073.0002 \cdot 142.0}, 2 \cdot 8000.0)$
 = 1936.3859 mm.

Input and Calculated Values:

Vessel Mean Radius	Rm	4073.00	mm.
Shell Thickness used in this Case	t	144.000	mm.
Stiffened Vessel Length per 4.15.6	L	82999.98	mm.
Distance from Saddle to Vessel tangent	a1 or a	8000.00	mm.
Saddle Width	b1 or b	750.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4073.00	mm.
Shell Allowable Stress used in Calculation		161.10	N./mm ²
Head Allowable Stress used in Calculation		161.10	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4545.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		788420.31	Kgf
Pressure used in Analysis	P	55.001	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	66.89	161.10
Long. Stress at Bottom of Midspan	90.87	161.10
Long. Stress at Top of Saddles	90.79	161.10
Long. Stress at Bottom of Saddles	72.28	161.10
Tangential Shear in Shell	11.86	128.88
Circ. Stress at Horn of Saddle	37.43	201.38
Circ. Compressive Stress in Shell	2.14	161.10

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = $Ftr (Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3.0(98103.7/4 + 0) \cdot 4545.0/7229.5811$
 = 46255.9 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(16865.43, 0) \cdot 4545.0/22333.9844$
 $= 3432.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(251181.36, 0) \cdot 4545.0/22333.9844$
 $= 51115.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(251181/4 + 0) \cdot 4545.0/7229.5811$
 $= 118432.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 669988 + \max(3432, 46256, 51116, 118432)$
 $= 788420.3 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1632.731(14.4 + \pi/4(8.288 \cdot 1.2)^2)$
 $= 165390.922 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	801185.62	Kgf
Transverse Shear Load	62795.34	Kgf
Longitudinal Shear Load	251181.36	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(8000.0/2 + 144.0 + 30.0) \sin(120.0/2), 2 \cdot 4073.0)$
 $= 7229.581 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/(3L))]$
 $= -788420 \cdot 8000.0 [1 - (1 - 8000.0/82999.98 + (4073.0^2 - 4073.0^2) / (2 \cdot 8000.0 \cdot 82999.98)) / (1 + (4 \cdot 4073.0) / (3 \cdot 82999.98))]$
 $= -957967.7 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 788420 \cdot 83000 / 4 ((1 + 2(4073^2 - 4073^2)/(83000^2)) / (1 + (4 \cdot 4073) / (3 \cdot 83000)) - 4 \cdot 8000 / 83000)$
 $= 9047859.0 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 55.001 \cdot 4073.0 / (2 \cdot 142.0) - 9047859 / (\pi \cdot 4073.0^2 \cdot 142.0)$
 $= 66.89 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 55.001 \cdot 4073.0 / (2 \cdot 142.0) + 9047859 / (\pi \cdot 4073.0^2 \cdot 142.0)$
 $= 90.87 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:
 $= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 55.001 \cdot 4073.0 / (2 \cdot 142.0) - 957967.7 / (0.1066 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$

$$= 90.79 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 55.001 \cdot 4073.0 / (2 \cdot 142.0) + 957967.7 / (0.1923 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 72.28 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 788420(82999.98 - 2 \cdot 8000.0) / (82999.98 + (4 \cdot 4073.0 / 3))$$

$$= 597351.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 597351 / (4073.0002 \cdot 142.0)$$

$$= 11.86 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(4073.0 \cdot 142.0)}$$

$$= 593.193 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 788420 \cdot 0.1 / (142.0(750.0 + 593.19 + 593.19))$$

$$= -2.14 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -788420 / (4 \cdot 142.0(750.0 + 593.193 + 593.193)) - 3 \cdot 0.0529 \cdot 788420 / (2 \cdot 142.0^2)$$

$$= -37.43 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(750.0 + 1.56 \cdot \sqrt{(4073.0 \cdot 142.0)}, 2 \cdot 8000.0)$$

$$= 1936.39 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7229.9995	mm.
Baseplate Thickness	Bpthk	130.0000	mm.
Baseplate Width	Bpwid	750.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	12	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	241.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(4002.0 \cdot 142.0)} + 900.0)^3) / 12$$

$$= 49534.836 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 49534.836 + 2947.92(173.236 - 71.0)^2$$

$$= 357657.875 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	2076.0	142.0	71.0	2947.9	20930232.0	0.358E+06
Wearplate	900.0	30.0	157.0	270.0	4239000.0	914.
Web	30.0	241.0	292.5	72.3	2114777.5	0.138E+05

BasePlate	750.0	130.0	478.0	975.0	46605016.0	0.919E+06
Totals	...	...	...	4265.2	73889024.0	0.129E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 73889024/4265.22$$

$$= 173.236 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 788420$$

$$= 160460.6250 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 11.9457 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 4545.0 - 4002.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 1235.373 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 198232.5781 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 26.0723 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (788420 + 12765) \cdot 750.0) / (4 \cdot 7230.0 \cdot 174.663)}$$

$$= 59.159 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (4000.0 + 144.0 + 30.0)$$

$$= 7229.580 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 7229.5801 / (12 - 1)$$

$$= 657.235 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 657.2346 \cdot 750.0/2$$

$$= 2464.630 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 801186 / (7230.0 \cdot 750.0)$$

$$= 14.775 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 2464.6 \cdot 14.78$$

$$= 36415.414 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



$$= 216.0 + 98.585$$

$$= 314.585 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 36415.4/314.5852$$

$$= 11.352 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	750.0	...	225.0	...	0.105E+06

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 750.0 - 30.0$$

$$= 720.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/314.585$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 750.0/2$$

$$= 375.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(105468.6/314.585)}$$

$$= 183.102 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 4545.0 - \cos(120.0/2)(4000.0 + 144.0 + 30.0) - 130.0$$

$$= 2328.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2)(4000.0 + 144.0 + 30.0)$$

$$= 7229.580 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 7230.0 - 50.8$$

$$= 7179.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (7229.58 - 7179.2) / 2$$

$$= 25.190 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{2328.0^2 + 25.19^2}$$

$$= 2328.136 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20219\text{E}+09/262.0)}$$

$$= 123.428$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2328.136/183.102$$

$$= 12.715$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 251181.4 / (2 \cdot 7230.0) \cdot 657.235 \cdot 2328.14 / 2$$

$$= 13290.048 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (12.715 < 123.4277) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.71)^2 / (2 \cdot 123.43^2)) 262 /$$

$$(5/3 + 3 \cdot (12.71) / (8 \cdot 123.43) - (12.71^3) / (8 \cdot 123.43^3))$$

$$= 152.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 11.35 / 152.83 + (13290.05 \cdot 375.0 / 1.0547E+10) / 174.66$$

$$= 0.340$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	720.0	0.0	216.0	0.0	0.105E+06
Web	657.2	30.0	0.0	197.2	0.0	148.
Totals	...	...	...	413.2	...	0.106E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 413.17$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 750.0 / 2$$

$$= 375.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 4545.0 - \sqrt{ (4174.0 + 30.0)^2 - (657.235/2)^2 } - 130.0$$

$$= 253.956 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 105609.8 / 413.17 }$$

$$= 159.878 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 253.956 / 159.878$$

$$= 1.588$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 251181.359 / (2 \cdot 7230.0)$$

$$= 17.371 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 17.371 \cdot 657.235 \cdot 253.956$$

$$= 2899.386 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2899.385 / 2816263$$

$$= 10.096 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (1.5884 < 123.4277) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (1.59)^2 / (2 \cdot 123.43^2)) 262 /$$

$$(5/3 + 3 \cdot (1.59) / (8 \cdot 123.43) - (1.59^3) / (8 \cdot 123.43^3))$$

= 156.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$
 = $17.61/156.73 + (2899.39 \cdot 375.0/105609.773) / 174.66$
 = 0.170

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	12
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	193.87 N./mm ²
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	17.2000 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BlArea	17.2000 cm ²
Saddle Load QO (Weight)	QO	682753.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	51115.8 Kg
Maximum Transverse Force	Ft	62795.3 Kg
Maximum Longitudinal Force	Fl	251181.4 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $Fl / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $251181.4 / (17.2 \cdot 12)$
 = 119.3 N./mm². Must be less than 129.3 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $62795.3 / (17.2 \cdot 12)$
 = 29.8 N./mm². Must be less than 129.3 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $Fl / (BoltShearAllowable \cdot Nbolts)$
 = $251181.36 / (129.27 \cdot 12.0)$
 = 15.8795 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
 = $4545.0 \cdot 62795.34 + 0.0$
 = 285410.47 Kg-m.

Eccentricity (e):

= $Rmom / QO$
 = $285410.47 / 682754$
 = 418.02 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 15.8795, 0.0]$
 = 15.8795 cm²

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.



ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

There are more than two saddles present in this model. The load on the saddle supports has been computed as the total weight divided by the number of saddles. Ensure that this is an appropriate assumption for this vessel model. It may be necessary to manually enter the bending moments at the saddle through the Saddle Input dialog. Also note that the dimension Ls is taken to be the distance between the adjoining saddles (first two or last two).

Note:

Wear Pad Width (900.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$ and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

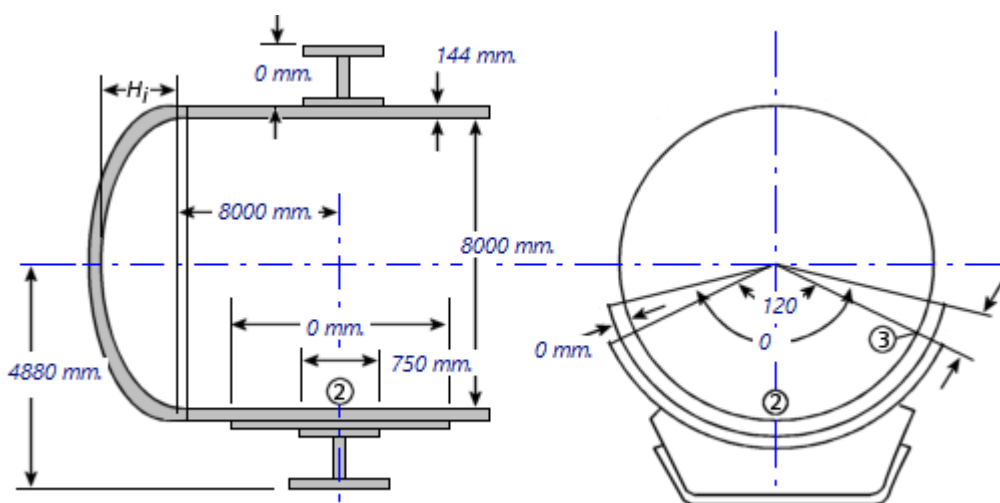
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(750.0 + 1.56 \cdot \sqrt{4073.0002 \cdot 142.0}, 2 \cdot 8000.0)$$

$$= 1936.3859 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	4073.00	mm.
Shell Thickness used in this Case	t	144.000	mm.
Stiffened Vessel Length per 4.15.6	L	82999.98	mm.
Distance from Saddle to Vessel tangent	a1 or a	8000.00	mm.
Saddle Width	b1 or b	750.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4073.00	mm.
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4880.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	1771094.88	Kgf
Pressure used in Analysis	P	74.841 bars

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	80.41	247.00

Long. Stress at Bottom of Midspan	134.27	247.00
Long. Stress at Top of Saddles	134.09	247.00
Long. Stress at Bottom of Saddles	92.51	247.00
Tangential Shear in Shell	26.64	138.58
Circ. Stress at Horn of Saddle	84.08	370.49
Circ. Compressive Stress in Shell	4.80	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3.0(32374.2/4 + 0) \cdot 4880.0/7229.5811$
 $= 16389.6 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl}, \text{wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(5565.59, 0) \cdot 4880.0/22333.0039$
 $= 1216.1 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 1754705 + \max(1216, 16390, 0, 0)$
 $= 1771094.9 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1632.731(14.4 + \pi/4(8.288 \cdot 1.2)^2)$
 $= 165390.922 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1785124.25	Kgf
Transverse Shear Load	8093.55	Kgf
Longitudinal Shear Load	5565.59	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(8000.0/2 + 144.0 + 30.0)\sin(120.0/2), 2 \cdot 4073.0)$
 $= 7229.581 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$
 $= -1771095 \cdot 8000.0 [1 - (1 - 8000.0/82999.98 + (4073.0^2 - 4073.0^2)/(2 \cdot 8000.0 \cdot 82999.98))/(1 + (4 \cdot 4073.0)/(3 \cdot 82999.98))]$
 $= -2151963.8 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$
 $= 1771095 \cdot 83000/4((1 + 2(4073^2 - 4073^2)/(83000^2))/(1 + (4 \cdot 4073)/(3 \cdot 83000)) - 4 \cdot 8000/83000)$
 $= 20324968.0 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$
 $= 74.841 \cdot 4073.0/(2 \cdot 142.0) - 20324968/(\pi \cdot 4073.0^2 \cdot 142.0)$
 $= 80.41 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot \text{Rm}/(2t) + \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$
 $= 74.841 \cdot 4073.0/(2 \cdot 142.0) + 20324968/(\pi \cdot 4073.0^2 \cdot 142.0)$

$$= 134.27 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 74.841 \cdot 4073.0 / (2 \cdot 142.0) - 2151963.8 / (0.1066 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 134.09 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 74.841 \cdot 4073.0 / (2 \cdot 142.0) + 2151963.8 / (0.1923 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 92.51 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 1771095(82999.98 - 2 \cdot 8000.0) / (82999.98 + (4 \cdot 4073.0 / 3))$$

$$= 1341880.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 1341880 / (4073.0002 \cdot 142.0)$$

$$= 26.64 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(4073.0 \cdot 142.0)}$$

$$= 593.193 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 1771095 \cdot 0.1 / (142.0(750.0 + 593.19 + 593.19))$$

$$= -4.80 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -1771095 / (4 \cdot 142.0(750.0 + 593.193 + 593.193)) - 3 \cdot 0.0529 \cdot 1771095 / (2 \cdot 142.0^2)$$

$$= -84.08 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(750.0 + 1.56 \cdot \sqrt{(4073.0 \cdot 142.0)}, 2 \cdot 8000.0)$$

$$= 1936.39 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7229.9995	mm.
Baseplate Thickness	Bpthk	130.0000	mm.
Baseplate Width	Bpwid	750.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	12	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	576.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(4002.0 \cdot 142.0)} + 900.0)^3) / 12$$

$$= 49534.836 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 49534.836 + 2947.92(257.427 - 71.0)^2$$

$$= 1074087.500 \text{ cm}^4$$

B	D	Y	A	AY	I + AD ²
mm.	mm.	mm.	cm ²	mm. ³	cm ⁴



Shell	2076.0	142.0	71.0	2947.9	20930232.0	0.107E+07
Wearplate	900.0	30.0	157.0	270.0	4239000.0	0.274E+05
Web	30.0	576.0	460.0	172.8	7948807.0	0.119E+06
BasePlate	750.0	130.0	813.0	975.0	79267520.0	0.302E+07
Totals	...	...	...	4365.7	112385552.0	0.424E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 0.11239E+09/4365.72$$

$$= 257.427 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1771095$$

$$= 360456.2188 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 24.9325 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 4880.0 - 4002.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 1570.373 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 566061.7500 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 33.6764 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1771095 + 14029)750.0) / (4 \cdot 7230.0 \cdot 173.33)}$$

$$= 88.645 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (4000.0 + 144.0 + 30.0)$$

$$= 7229.580 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 7229.5801 / (12 - 1)$$

$$= 657.235 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 657.2346 \cdot 750.0/2$$

$$= 2464.630 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1785124 / (7230.0 \cdot 750.0)$$

$$= 32.921 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 2464.6 \cdot 32.92$$

$$= 81137.305 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 216.0 + 98.585 \\
 &= 314.585 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/Ar \\
 &= 81137.3/314.5852 \\
 &= 25.294 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	750.0	...	225.0	...	0.105E+06

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 750.0 - 30.0 \\
 &= 720.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0/314.585 \\
 &= 0.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 750.0/2 \\
 &= 375.000 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\
 &= \sqrt{105468.6/314.585} \\
 &= 183.102 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 4880.0 - \cos(120.0/2)(4000.0 + 144.0 + 30.0) - 130.0 \\
 &= 2663.000 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(120.0/2)(4000.0 + 144.0 + 30.0) \\
 &= 7229.580 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= \text{Baseplate length} - \text{clearance} \\
 &= 7230.0 - 50.8 \\
 &= 7179.200 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd)/2 \\
 &= (7229.58 - 7179.2)/2 \\
 &= 25.190 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \sqrt{Lv^2 + Dd^2} \\
 &= \sqrt{2663.0^2 + 25.19^2} \\
 &= 2663.119 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}} \\
 &= \sqrt{2 \cdot \pi^2 \cdot 0.20219\text{E}+09/260.0} \\
 &= 123.902
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2663.119/183.102$$

$$= 14.544$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 5565.6 / (2 \cdot 7230.0) \cdot 657.235 \cdot 2663.12/2$$

$$= 336.847 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (14.5445 < 123.9015) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (14.54)^2 / (2 \cdot 123.9^2)) 260 / (5/3 + 3 \cdot (14.54) / (8 \cdot 123.9) - (14.54^3) / (8 \cdot 123.9^3))$$

$$= 151 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 25.29/150.95 + (336.85 \cdot 375.0 / .10547E+10) / 173.33$$

$$= 0.174$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	720.0	0.0	216.0	0.0	0.105E+06
Web	657.2	30.0	0.0	197.2	0.0	148.
Totals	...	...	...	413.2	...	0.106E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/413.17$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 750.0/2$$

$$= 375.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 4880.0 - \sqrt{ (4174.0 + 30.0)^2 - (657.235/2)^2 } - 130.0$$

$$= 588.956 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 105609.8/413.17 }$$

$$= 159.878 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 588.956/159.878$$

$$= 3.684$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 5565.593 / (2 \cdot 7230.0)$$

$$= 0.385 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.385 \cdot 657.235 \cdot 588.956$$

$$= 148.989 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 148.989/2816263$$

$$= 0.519 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (3.6838 < 123.9015) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (3.68)^2 / (2 \cdot 123.9^2)) 260 /$$

$$(5/3 + 3 \cdot (3.68) / (8 \cdot 123.9) - (3.68^3) / (8 \cdot 123.9^3))$$

$$= 154.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 39.24/154.89 + (148.99 \cdot 375.0 / 105609.773) / 173.33$$

$$= 0.256$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	12
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	17.2000 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	17.2000 cm ²
Saddle Load QO (Weight)	QO	1768734.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1216.1 Kg
Maximum Transverse Force	Ft	8093.6 Kg
Maximum Longitudinal Force	Fl	5565.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5565.6 / (17.2 \cdot 12)$$

$$= 2.6 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8093.6 / (17.2 \cdot 12)$$

$$= 3.8 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 5565.59 / (103.2 \cdot 12.0)$$

$$= 0.4407 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 4880.0 \cdot 8093.55 + 0.0$$

$$= 39497.32 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 39497.32 / 1768735$$

$$= 22.33 \text{ mm. < Bplen/6 --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.4407, 0.0]$$

$$= 0.4407 \text{ cm}^2$$



Note:
 Wear Pad Width (900.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$
 $= \min(750.0 + 1.56 \cdot \sqrt{ 4073.0002 \cdot 142.0 }, 2 \cdot 8000.0)$
 $= 1936.3859 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	4073.00	mm.
Shell Thickness used in this Case	t	144.000	mm.
Stiffened Vessel Length per 4.15.6	L	82999.98	mm.
Distance from Saddle to Vessel tangent	a1 or a	8000.00	mm.
Saddle Width	b1 or b	750.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	4073.00	mm.
Shell Allowable Stress used in Calculation		247.00	N./mm ²
Head Allowable Stress used in Calculation		247.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	4545.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	1769969.75	Kgf
Pressure used in Analysis	P 74.841	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	80.42	247.00
Long. Stress at Bottom of Midspan	134.26	247.00
Long. Stress at Top of Saddles	134.07	247.00
Long. Stress at Bottom of Saddles	92.52	247.00
Tangential Shear in Shell	26.62	138.58
Circ. Stress at Horn of Saddle	84.03	370.49
Circ. Compressive Stress in Shell	4.80	247.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(32374.2/4 + 0) \cdot 4545.0/7229.5811$
 $= 15264.5 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(5565.59, 0) \cdot 4545.0/22333.9844$
 $= 1132.6 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 1754705 + \max(1133, 15264, 0, 0)$
 $= 1769969.8 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1632.731(14.4 + \pi/4(8.288 \cdot 1.2)^2)$
 $= 165390.922 \text{ N}$

Summary of Loads at the Base of this Saddle:



Vertical Load (including saddle weight)	1782735.00	Kgf
Transverse Shear Load	8093.55	Kgf
Longitudinal Shear Load	5565.59	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(8000.0/2 + 144.0 + 30.0)\sin(120.0/2), 2 \cdot 4073.0)$$

$$= 7229.581 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -1769970 \cdot 8000.0 [1 - (1 - 8000.0/82999.98 + (4073.0^2 - 4073.0^2)/(2 \cdot 8000.0 \cdot 82999.98))/(1 + (4 \cdot 4073.0)/(3 \cdot 82999.98))]$$

$$= -2150596.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 1769970 \cdot 83000/4((1 + 2(4073^2 - 4073^2)/(83000^2))/(1 + (4 \cdot 4073)/(3 \cdot 83000)) - 4 \cdot 8000/83000)$$

$$= 20312054.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 74.841 \cdot 4073.0/(2 \cdot 142.0) - 20312054/(\pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 80.42 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 74.841 \cdot 4073.0/(2 \cdot 142.0) + 20312054/(\pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 134.26 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 74.841 \cdot 4073.0/(2 \cdot 142.0) - 2150596.5/(0.1066 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 134.07 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 74.841 \cdot 4073.0/(2 \cdot 142.0) + 2150596.5/(0.1923 \cdot \pi \cdot 4073.0^2 \cdot 142.0)$$

$$= 92.52 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 1769970(82999.98 - 2 \cdot 8000.0)/(82999.98 + (4 \cdot 4073.0/3))$$

$$= 1341027.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1341028/(4073.0002 \cdot 142.0)$$

$$= 26.62 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(4073.0 \cdot 142.0)}$$

$$= 593.193 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 1769970 \cdot 0.1/(142.0(750.0 + 593.19 + 593.19))$$

$$= -4.80 \text{ N./mm}^2$$



Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q/(2 \cdot t^2)$$

$$= -1769970/(4 \cdot 142.0(750.0 + 593.193 + 593.193)) -$$

$$3 \cdot 0.0529 \cdot 1769970/(2 \cdot 142.0^2)$$

$$= -84.03 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(750.0 + 1.56 \cdot \sqrt{4073.0 \cdot 142.0}, 2 \cdot 8000.0)$$

$$= 1936.39 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	7229.9995	mm.
Baseplate Thickness	Bpthk	130.0000	mm.
Baseplate Width	Bpwid	750.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	12	
Rib Thickness	Ribtk	30.0000	mm.
Web Thickness	Webtk	30.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	241.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{4002.0 \cdot 142.0} + 900.0)142.0^3)/12$$

$$= 49534.836 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 49534.836 + 2947.92(173.236 - 71.0)^2$$

$$= 357657.875 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	2076.0	142.0	71.0	2947.9	20930232.0	0.358E+06
Wearplate	900.0	30.0	157.0	270.0	4239000.0	914.
Web	30.0	241.0	292.5	72.3	2114777.5	0.138E+05
BasePlate	750.0	130.0	478.0	975.0	46605016.0	0.919E+06
Totals	...	...	...	4265.2	73889024.0	0.129E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 73889024/4265.22$$

$$= 173.236 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2)/(\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.204 \cdot 1769970$$

$$= 360227.2188 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 26.8176 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians})$$

$$= 4545.0 - 4002.0 \cdot \sin(120.0/2)/1.0472$$

$$= 1235.373 \text{ mm.}$$

$$\text{Bending Moment, } M = F_h \cdot d = 445023.6562 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 58.5312 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(1769970 + 12765)750.0 / (4 \cdot 7230.0 \cdot 173.33)}$$

$$= 88.586 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (4000.0 + 144.0 + 30.0)$$

$$= 7229.580 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 7229.5801 / (12 - 1)$$

$$= 657.235 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 657.2346 \cdot 750.0/2$$

$$= 2464.630 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1782735 / (7230.0 \cdot 750.0)$$

$$= 32.877 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 2464.6 \cdot 32.88$$

$$= 81028.711 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 216.0 + 98.585$$

$$= 314.585 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 81028.7 / 314.5852$$

$$= 25.260 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	30.0	750.0	...	225.0	...	0.105E+06

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 750.0 - 30.0$$

$$= 720.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 314.585$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 750.0/2$$

$$= 375.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{105468.6/314.585}$$

$$= 183.102 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 4545.0 - \cos(120.0/2) (4000.0 + 144.0 + 30.0) - 130.0$$

$$= 2328.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (4000.0 + 144.0 + 30.0)$$

$$= 7229.580 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 7230.0 - 50.8$$

$$= 7179.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (7229.58 - 7179.2) / 2$$

$$= 25.190 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{2328.0^2 + 25.19^2}$$

$$= 2328.136 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.20219\text{E}+09 / 260.0}$$

$$= 123.902$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 2328.136 / 183.102$$

$$= 12.715$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 5565.6 / (2 \cdot 7230.0) \cdot 657.235 \cdot 2328.14 / 2$$

$$= 294.476 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($12.715 < 123.9015$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (12.71)^2 / (2 \cdot 123.9^2)) 260 /$$

$$(5/3 + 3 \cdot (12.71) / (8 \cdot 123.9) - (12.71^3) / (8 \cdot 123.9^3))$$

$$= 151.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 25.26 / 151.69 + (294.48 \cdot 375.0 / 1.0547\text{E}+10) / 173.33$$

$$= 0.172$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	30.0	720.0	0.0	216.0	0.0	0.105E+06
Web	657.2	30.0	0.0	197.2	0.0	148.
Totals	...	...	...	413.2	...	0.106E+06

Distance to Centroid from Datum [ytot]:

$$= AY / A$$



= 0.0/413.17
 = 0.000 mm.

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 750.0/2
 = 375.000 mm.

Length of Inner Rib [L]:
 = Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
 = 4545.0 - $\sqrt{(4174.0 + 30.0)^2 - (657.235/2)^2}$ - 130.0
 = 253.956 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{105609.8/413.17}$
 = 159.878 mm.

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 253.956/159.878
 = 1.588

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = 5565.593/(2 • 7230.0)
 = 0.385 Kg/mm.

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = 0.385 • 657.235 • 253.956
 = 64.244 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 64.244/2816263
 = 0.224 N./mm²

Compressive Allowable, KL/r < Cc (1.5884 < 123.9015) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (1.59)^2 / (2 \cdot 123.9^2)) 260 /$
 $(5/3 + 3 \cdot (1.59) / (8 \cdot 123.9) - (1.59^3) / (8 \cdot 123.9^3))$
 = 155.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = Sc/Sca + (Mbase,c • C1/I)/Sba
 = 39.19/155.54 + (64.24 • 375.0/105609.773)/173.33
 = 0.253

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	12
Total Number of Bolts in Tension/Baseplate	Nbt	8
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	0.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	17.2000 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BlArea	17.2000 cm ²
Saddle Load Q0 (Weight)	Q0	1767470.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1132.6 Kg
Maximum Transverse Force	Ft	8093.6 Kg
Maximum Longitudinal Force	Fl	5565.6 Kg
Saddle Bolted to Steel Foundation	No	No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = FI / (Bolt Area • Number of Bolts)
 = 5565.6/(17.2 • 12)
 = 2.6 N./mm². Must be less than 103.2 N./mm².



Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $8093.6 / (17.2 \cdot 12)$
= 3.8 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_t / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $5565.59 / (103.2 \cdot 12.0)$
= 0.4407 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $4545.0 \cdot 8093.55 + 0.0$
= 36785.93 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $36785.93 / 1767471$
= 20.81 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 0.4407, 0.0]$
= 0.4407 cm²

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: M1

From: 30

Pressure for Reinforcement Calculations	P	55.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4500.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

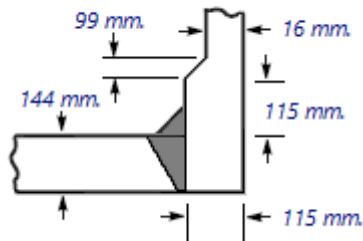
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	115.0000	mm.
Height of Beveled Transition	L'	99.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		115.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	432.8000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(55.0/(153.141 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 11.2561 + 3.0 + 0.0 = 14.2561 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln((2 \cdot 13.0 + 615.6) / 615.6) - 0.0$$

$$= 63.347 - 0.0 = 63.3473 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 115.0 - 3.0$$

$$= 112.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 3.0$$

$$= 13.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4002.0 \cdot 142.0}, 2 \cdot 307.8)$$

$$= 615.600 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t_e, \sqrt{R_n \cdot t_n}]$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{307.8 \cdot 112.0}$$

$$= 185.671 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 432.8$$

$$= 432.800 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[185.671, 432.8, 1136.0] + 142.0$$

$$= 327.671 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 615.6)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right)$$

$$= \min\left(12, \left(2 \cdot 307.8 + 112.0\right) / \left(\sqrt{\left(8004.0 + 142.0\right) \cdot 142.0}\right)\right)$$

$$= 0.677$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right)$$

$$= 142.0 \cdot 615.6 \cdot \max\left(\left(0.677/5\right)^{0.85}, 1\right)$$

$$= 874.152 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 287.84 + 54.179$$

$$= 342.020 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 874.152 + 0.951(342.02 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 1199.775 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 112.0 / \ln[1 + 112.0/307.8]$$

$$= 360.908 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 55.0 \cdot 360.908 \cdot 327.671$$

$$= 66327.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 55.0 \cdot 4072.584(615.6 + 112.0)$$



= 1661970.1 Kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $55.0 \cdot 4072.584 \cdot 307.8$
 = 703070.9 Kgf

Area Resisting Internal Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(1661970 + 703071 + 66327.7) / 55.0$
 = 43350.1 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$
 = $241.65 / (2 \cdot 43350.125 / 1199.775 - 4072.584 / 142.0)$
 = 55.4 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
 = $161.1 [142.0 / 4072.584]$
 = 56.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(55.442, 56.168)$
 = 55.442 bars

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / AT$
 = $(66327.695 + 1661970 + 703071) / 1199.775$
 = 198.737 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / t_{eff}$
 = $55.0 \cdot 4072.584 / 142.0$
 = 157.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 198.737 - 157.751, 157.751)$
 = 239.7 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	239.72	N./mm ²
Maximum Allowable Working Pressure	Pmax	55.44	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= $(R_{nc} + t_n) / R_{nc}$
 = $(307.8 + 112.0) / 307.8$
 = 1.364 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi / 2 \cdot (R_n + t_n)$
 = $\pi / 2 \cdot (307.8 + 112.0)$
 = 659.420 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 112.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot ky^2)$
 = $\min(703071 \cdot 1.36, 1.5 \cdot 153.1(342.02 + 0.0), \pi / 4 \cdot 55.0 \cdot 307.8^2 \cdot 1.36^2)$
 = 77631.008 Kgf



Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 77631.008 / (659.42(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
 $= 52.324 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -2 °C
 User entered Minimum Design Temperature : 0 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 8 °C
 Temperature reduction per Graph 3.13 : 11 °C
 The computed stress ratio is : 0.803

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (10.434 \cdot 1.0) / (16.0 - 3.0 - 0.0)$
 $= 0.803$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -2 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 22.0901 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 598.8902 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N2

From: 40

Pressure for Reinforcement Calculations	P	55.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	2650.0000	mm.
Distance from Bottom/Left Tangent		8499.98	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

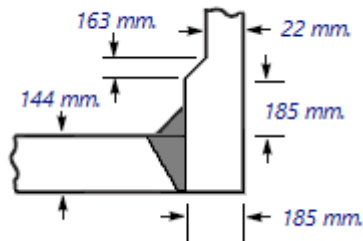
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		42.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm.
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	185.0000	mm.
Height of Beveled Transition	L'	163.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		185.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	356.6000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	144.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	42.000	in.
Actual Thickness Used in Calculation	0.866	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1070.8 \cdot (\exp(55.0/(153.141 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 19.5794 + 2.0 + 0.0 = 21.5794 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln((2 \cdot 20.0 + 1070.8) / 1070.8) - 0.0$$

$$= 56.16 - 0.0 = 56.1604 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.340 mm
Min. required shell thickness, internal p + c	141.011 mm.
Nozzle neck minimum required thickness	: 10.340 mm.

Minimum nozzle neck thickness 21.996 mm., is greater than required thickness 10.340 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 185.0 - 2.0$$

$$= 183.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 22.0 - 2.0$$

$$= 20.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$



$$= \min(\sqrt{(4002.0 \cdot 142.0)}, 2 \cdot 535.4)$$

$$= 608.600 \text{ mm.}$$

Note: LR is governed by the user entered diameter limit.

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{(535.4 \cdot 183.0)}$$

$$= 313.015 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 356.6$$

$$= 356.600 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[313.015, 356.6, 1136.0] + 142.0$$

$$= 455.015 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 608.6)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 535.4 + 183.0) / (\sqrt{(8004.0 + 142.0) \cdot 142.0}))$$

$$= 1.166$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 142.0 \cdot 608.6 \cdot \max((1.166/5)^{0.85}, 1)$$

$$= 864.212 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 598.41 + 152.328$$

$$= 750.738 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 864.212 + 0.951(750.738 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 1578.362 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 183.0 / \ln[1 + 183.0/535.4]$$

$$= 622.423 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

= 241.7 N./mm²

Force acting on the Nozzle [fN]:

= P • Rxn • LH
 = 55.0 • 622.423 • 455.015
 = 158844.2 Kgf

Force acting on the Shell [fS]:

= P • Rxs(LR + tn)
 = 55.0 • 4072.584(608.6 + 183.0)
 = 1808157.8 Kgf

Discontinuity Force from Internal Pressure [fY]:

= P • Rxs • Rnc
 = 55.0 • 4072.584 • 535.4
 = 1222950.5 Kgf

Area Resisting Internal Pressure [Ap]:

= (fS + fY + fN)/P
 = (1808158 + 1222951 + 158844.2)/55.0
 = 56875.3 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= Sallow / (2 • Ap/AT - Rxs/teff)
 = 241.65 / (2 • 56875.309/1578.362 - 4072.584/142.0)
 = 55.7 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= S[t/Rxs]
 = 161.1[142.0/4072.584]
 = 56.2 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
 = min(55.691, 56.168)
 = 55.691 bars

Average Primary Membrane Stress [σAvg]:

= (fN + fS + fY) / AT
 = (158844.203 + 1808158 + 1222951)/1578.362
 = 198.201 N./mm²

General Primary Membrane Stress [σCirc]:

= P • Rxs / teff
 = 55.0 • 4072.584/142.0
 = 157.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σAvg - σCirc, σCirc)
 = max(2 • 198.201 - 157.751, 157.751)
 = 238.7 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	238.65	N./mm ²
Maximum Allowable Working Pressure	Pmax	55.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= (Rnc + tn) / Rnc
 = (535.4 + 183.0)/535.4
 = 1.342 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= π/2 • (Rn + tn)
 = π/2 • (535.4 + 183.0)
 = 1128.460 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 183.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(1222951 \cdot 1.34, 1.5 \cdot 153.1(750.738 + 0.0), \pi/4 \cdot 55.0 \cdot 535.4^2 \cdot 1.34^2)$
= 227343.766 Kgf

Weld Stress Value [τ]:

= $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $227343.766 / (1128.46(0.49 \cdot 7.071 + 0.6 \cdot 142.0 + 0.49 \cdot 0.0))$
= 22.283 < or = to 161.100 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -73 °C

The Drop for this Nozzle is : 65.0410 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 565.6411 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N1

From: 40

Pressure for Reinforcement Calculations	P	55.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1300.0000	mm.
Distance from Bottom/Left Tangent		6499.98	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

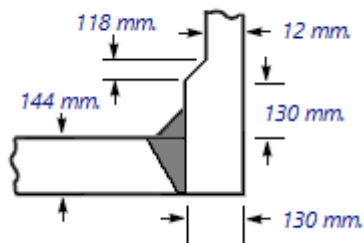
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	130.0000	mm.
Height of Beveled Transition	L'	118.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		130.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	445.5000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	144.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	20.000	in.
Actual Thickness Used in Calculation	0.472	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 512.0 \cdot (\exp(55.0/(153.141 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 9.3618 + 2.0 + 0.0 = 11.3618 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln\left(\frac{2 \cdot t + D}{D}\right) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln\left(\frac{2 \cdot 10.0 + 512.0}{512.0}\right) - 0.0$$

$$= 58.679 - 0.0 = 58.6786 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.340 mm
Min. required shell thickness, internal p + c :	141.011 mm.
Nozzle neck minimum required thickness :	10.340 mm.

Minimum nozzle neck thickness 11.989 mm., is greater than required thickness 10.340 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 130.0 - 2.0$$

$$= 128.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 12.0 - 2.0$$

$$= 10.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$



$$= \min(\sqrt{(4002.0 \cdot 142.0)}, 2 \cdot 256.0)$$

$$= 268.000 \text{ mm.}$$

Note: LR is governed by the user entered diameter limit.

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{(256.0 \cdot 128.0)}$$

$$= 181.019 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 445.5$$

$$= 445.500 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[181.019, 445.5, 1136.0] + 142.0$$

$$= 323.019 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 268.0)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})))$$

$$= \min(12, (2 \cdot 256.0 + 128.0) / (\sqrt{(8004.0 + 142.0) \cdot 142.0}))$$

$$= 0.595$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 142.0 \cdot 268.0 \cdot \max((0.595/5)^{0.85}, 1)$$

$$= 380.560 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A_{2A} + A_{2B}$$

$$= 348.16 + 52.29$$

$$= 400.450 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg_1^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A_1 + frn(A_2 + A_3) + A_{41} + A_{42} + A_{43}$$

$$= 380.56 + 0.951(400.45 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 761.727 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 128.0 / \ln[1 + 128.0/256.0]$$

$$= 315.687 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$



$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 55.0 \cdot 315.687 \cdot 323.019$$

$$= 57193.3 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 55.0 \cdot 4072.584 (268.0 + 128.0)$$

$$= 904535.6 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 55.0 \cdot 4072.584 \cdot 256.0$$

$$= 584750.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (904536 + 584750 + 57193.3) / 55.0$$

$$= 27573.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 241.65 / (2 \cdot 27572.977 / 761.727 - 4072.584 / 142.0)$$

$$= 55.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 161.1 [142.0 / 4072.584]$$

$$= 56.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(55.274, 56.168)$$

$$= 55.274 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (57193.34 + 904536 + 584750) / 761.727$$

$$= 199.101 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 55.0 \cdot 4072.584 / 142.0$$

$$= 157.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 199.101 - 157.751, 157.751)$$

$$= 240.5 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	240.45	N./mm ²
Maximum Allowable Working Pressure	Pmax	55.27	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (256.0 + 128.0) / 256.0$$

$$= 1.500 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lr]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (256.0 + 128.0)$$

$$= 603.186 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 128.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(584750 \cdot 1.5, 1.5 \cdot 153.1(400.45 + 0.0), \pi/4 \cdot 55.0 \cdot 256.0^2 \cdot 1.5^2)$
= 64955.023 Kgf

Weld Stress Value [τ]:

= $fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $64955.023 / (603.186(0.49 \cdot 7.071 + 0.6 \cdot 142.0 + 0.49 \cdot 0.0))$
= 11.911 $\leq$ or = to 161.100 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-3 °C
User entered Minimum Design Temperature	:	0 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	3 °C
Temperature reduction per Graph 3.13	:	7 °C
The computed stress ratio is	:	0.868

The computed thickness ratio [sr]:

= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(8.678 \cdot 1.0) / (12.0 - 2.0 - 0.0)$
= 0.868

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -3 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 18.4745 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 607.9745 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: M2

From: 160

Pressure for Reinforcement Calculations	P	55.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		43500.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

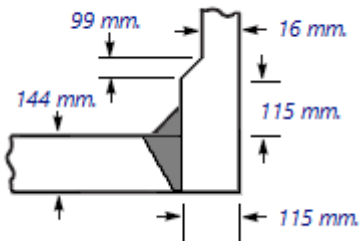
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	115.0000	mm.
Height of Beveled Transition	L'	99.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		115.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	432.8000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(55.0/(153.141 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 11.2561 + 3.0 + 0.0 = 14.2561 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln\left(\frac{2 \cdot t + D}{D}\right) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln\left(\frac{2 \cdot 13.0 + 615.6}{615.6}\right) - 0.0$$

$$= 63.347 - 0.0 = 63.3473 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 115.0 - 3.0$$

$$= 112.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 3.0$$

$$= 13.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4002.0 \cdot 142.0}, 2 \cdot 307.8)$$

$$= 615.600 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t_e, \sqrt{R_n \cdot t_n}]$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{307.8 \cdot 112.0}$$

$$= 185.671 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 432.8$$

$$= 432.800 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[185.671, 432.8, 1136.0] + 142.0$$

$$= 327.671 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 615.6)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right)$$

$$= \min\left(12, \left(2 \cdot 307.8 + 112.0\right) / \left(\sqrt{\left(8004.0 + 142.0\right) \cdot 142.0}\right)\right)$$

$$= 0.677$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right)$$

$$= 142.0 \cdot 615.6 \cdot \max\left(\left(0.677/5\right)^{0.85}, 1\right)$$

$$= 874.152 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 287.84 + 54.179$$

$$= 342.020 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 874.152 + 0.951(342.02 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 1199.775 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 112.0 / \ln[1 + 112.0/307.8]$$

$$= 360.908 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 55.0 \cdot 360.908 \cdot 327.671$$

$$= 66327.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 55.0 \cdot 4072.584(615.6 + 112.0)$$



= 1661970.1 Kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $55.0 \cdot 4072.584 \cdot 307.8$
 = 703070.9 Kgf

Area Resisting Internal Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(1661970 + 703071 + 66327.7) / 55.0$
 = 43350.1 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$
 = $241.65 / (2 \cdot 43350.125 / 1199.775 - 4072.584 / 142.0)$
 = 55.4 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
 = $161.1 [142.0 / 4072.584]$
 = 56.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(55.442, 56.168)$
 = 55.442 bars

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / AT$
 = $(66327.695 + 1661970 + 703071) / 1199.775$
 = 198.737 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / t_{eff}$
 = $55.0 \cdot 4072.584 / 142.0$
 = 157.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 198.737 - 157.751, 157.751)$
 = 239.7 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	239.72	N./mm ²
Maximum Allowable Working Pressure	Pmax	55.44	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= $(R_{nc} + t_n) / R_{nc}$
 = $(307.8 + 112.0) / 307.8$
 = 1.364 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi / 2 \cdot (R_n + t_n)$
 = $\pi / 2 \cdot (307.8 + 112.0)$
 = 659.420 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 112.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot ky^2)$
 = $\min(703071 \cdot 1.36, 1.5 \cdot 153.1(342.02 + 0.0), \pi / 4 \cdot 55.0 \cdot 307.8^2 \cdot 1.36^2)$
 = 77631.008 Kgf



Weld Stress Value [τ]:
 $= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 77631.008 / (659.42(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
 $= 52.324 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -2 °C
 User entered Minimum Design Temperature : 0 °C
 The Curve for this Material is : A
 From Figure 3.8M- with PWHT, the MDMT is : 8 °C
 Temperature reduction per Graph 3.13 : 11 °C
 The computed stress ratio is : 0.803

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (10.434 \cdot 1.0) / (16.0 - 3.0 - 0.0)$
 $= 0.803$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -2 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -46 °C

The Drop for this Nozzle is : 22.0901 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 598.8902 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K2

From: 270

Pressure for Reinforcement Calculations	P	55.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		74499.98	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

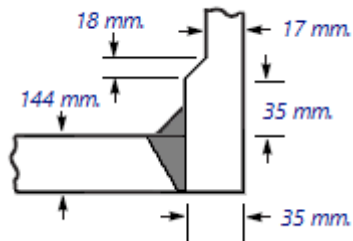
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	35.0000	mm.
Height of Beveled Transition	L'	18.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	562.8480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	144.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.669	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 54.8 \cdot (\exp(55.0/(153.141 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 1.002 + 2.0 + 0.0 = 3.002 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 54.8) / 54.8) - 0.0$$

$$= 668.584 - 0.0 = 668.584 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.800 mm
Min. required shell thickness, internal p + c	141.011 mm.
Nozzle neck minimum required thickness	6.800 mm.

Minimum nozzle neck thickness 16.993 mm., is greater than required thickness 6.800 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 35.0 - 2.0$$

$$= 33.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 17.0 - 2.0$$

$$= 15.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{(4002.0 \cdot 142.0)}, 2 \cdot 27.4)$$

$$= 54.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{(27.4 \cdot 33.0)}$$

$$= 30.070 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 562.848$$

$$= 562.848 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[30.07, 562.848, 1136.0] + 142.0$$

$$= 172.070 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 54.8)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 27.4 + 33.0) / (\sqrt{(8004.0 + 142.0) \cdot 142.0}))$$

$$= 0.082$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 142.0 \cdot 54.8 \cdot \max((0.082/5)^{0.85}, 1)$$

$$= 77.816 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 33.0 \cdot 172.07$$

$$= 56.783 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 77.816 + 0.951(56.783 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 132.294 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 33.0 / \ln[1 + 33.0/27.4]$$

$$= 41.749 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 55.0 \cdot 41.749 \cdot 172.07$$

$$= 4029.1 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 55.0 \cdot 4072.584 (54.8 + 33.0)$$

$$= 200551.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 55.0 \cdot 4072.584 \cdot 27.4$$

$$= 62586.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (200551.1 + 62586.6 + 4029.1) / 55.0$$

$$= 4763.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 241.65 / (2 \cdot 4763.454 / 132.294 - 4072.584 / 142.0)$$

$$= 55.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 161.1 [142.0 / 4072.584]$$

$$= 56.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(55.762, 56.168)$$

$$= 55.762 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (4029.092 + 200551.078 + 62586.559) / 132.294$$

$$= 198.048 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 55.0 \cdot 4072.584 / 142.0$$

$$= 157.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.048 - 157.751, 157.751)$$

$$= 238.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	238.35	N./mm ²
Maximum Allowable Working Pressure	Pmax	55.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (27.4 + 33.0) / 27.4$$

$$= 2.204 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (27.4 + 33.0)$$

$$= 94.876 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 * 33.000, 6.000)$ Passed

Weld Load Value [fwelds]:
 $= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(62587 \cdot 2.2, 1.5 \cdot 153.1(56.783 + 0.0), \pi/4 \cdot 55.0 \cdot 27.4^2 \cdot 2.2^2)$
 $= 1607.031 \text{ Kgf}$

Weld Stress Value [τ]:
 $= \text{fwelds} / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 1607.031 / (94.876(0.49 \cdot 7.071 + 0.6 \cdot 142.0 + 0.49 \cdot 0.0))$
 $= 1.873 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
 Computed Minimum Design Metal Temperature : -104 °C
 User entered Minimum Design Temperature : 0 °C
 The Curve for this Material is : A

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
 The computed stress ratio is : 0.062

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (0.929 \cdot 1.0) / (17.0 - 2.0 - 0.0)$
 $= 0.062$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification : -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction : -46 °C

The Drop for this Nozzle is : 0.4558 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 707.3039 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: K1

From: 270

Pressure for Reinforcement Calculations	P	55.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		76499.98	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

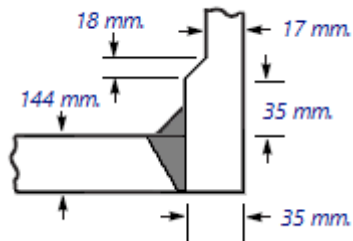
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	35.0000	mm.
Height of Beveled Transition	L'	18.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	562.8480	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	144.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: K1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.669	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 54.8 \cdot (\exp(55.0/(153.141 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 1.002 + 2.0 + 0.0 = 3.002 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 54.8) / 54.8) - 0.0$$

$$= 668.584 - 0.0 = 668.584 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.800 mm
Min. required shell thickness, internal p + c	141.011 mm.
Nozzle neck minimum required thickness	6.800 mm.

Minimum nozzle neck thickness 16.993 mm., is greater than required thickness 6.800 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 35.0 - 2.0$$

$$= 33.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 17.0 - 2.0$$

$$= 15.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$



$$= \min(\sqrt{(4002.0 \cdot 142.0)}, 2 \cdot 27.4)$$

$$= 54.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{(27.4 \cdot 33.0)}$$

$$= 30.070 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 562.848$$

$$= 562.848 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[30.07, 562.848, 1136.0] + 142.0$$

$$= 172.070 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 54.8)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 27.4 + 33.0) / (\sqrt{(8004.0 + 142.0) \cdot 142.0}))$$

$$= 0.082$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 142.0 \cdot 54.8 \cdot \max((0.082/5)^{0.85}, 1)$$

$$= 77.816 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 33.0 \cdot 172.07$$

$$= 56.783 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 77.816 + 0.951(56.783 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 132.294 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 33.0 / \ln[1 + 33.0/27.4]$$

$$= 41.749 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 55.0 \cdot 41.749 \cdot 172.07$$

$$= 4029.1 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 55.0 \cdot 4072.584 (54.8 + 33.0)$$

$$= 200551.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 55.0 \cdot 4072.584 \cdot 27.4$$

$$= 62586.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (200551.1 + 62586.6 + 4029.1) / 55.0$$

$$= 4763.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 241.65 / (2 \cdot 4763.454 / 132.294 - 4072.584 / 142.0)$$

$$= 55.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 161.1 [142.0 / 4072.584]$$

$$= 56.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(55.762, 56.168)$$

$$= 55.762 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (4029.092 + 200551.078 + 62586.559) / 132.294$$

$$= 198.048 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 55.0 \cdot 4072.584 / 142.0$$

$$= 157.8 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.048 - 157.751, 157.751)$$

$$= 238.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	238.35	N./mm ²
Maximum Allowable Working Pressure	Pmax	55.76	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (27.4 + 33.0) / 27.4$$

$$= 2.204 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (27.4 + 33.0)$$

$$= 94.876 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 33.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(62587 \cdot 2.2, 1.5 \cdot 153.1(56.783 + 0.0), \pi/4 \cdot 55.0 \cdot 27.4^2 \cdot 2.2^2)$
 $= 1607.031 \text{ Kgf}$

Weld Stress Value [τ]:

$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 $= 1607.031 / (94.876(0.49 \cdot 7.071 + 0.6 \cdot 142.0 + 0.49 \cdot 0.0))$
 $= 1.873 < \text{or} = \text{to } 161.100 \text{ Weld Stress Passed}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : 0 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240 , the MDMT is : -104 °C
The computed stress ratio is : 0.062

The computed thickness ratio [sr]:

$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (0.929 \cdot 1.0) / (17.0 - 2.0 - 0.0)$
 $= 0.062$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification : -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction : -46 °C

The Drop for this Nozzle is : 0.4558 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 707.3039 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: M3

From: 280

Pressure for Reinforcement Calculations	P	55.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		78499.98	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

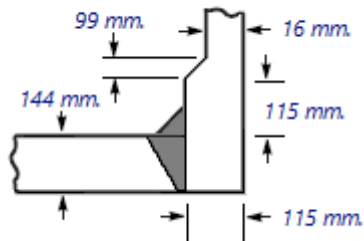
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	115.0000	mm.
Height of Beveled Transition	L'	99.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		115.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Outside Projection	ho	432.8000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	31.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	24.000	in.
Actual Thickness Used in Calculation	0.630	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 615.6 \cdot (\exp(55.0/(153.141 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 11.2561 + 3.0 + 0.0 = 14.2561 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln\left(\frac{2 \cdot t + D}{D}\right) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln\left(\frac{2 \cdot 13.0 + 615.6}{615.6}\right) - 0.0$$

$$= 63.347 - 0.0 = 63.3473 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 115.0 - 3.0$$

$$= 112.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 3.0$$

$$= 13.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4002.0 \cdot 142.0}, 2 \cdot 307.8)$$

$$= 615.600 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{307.8 \cdot 112.0}$$

$$= 185.671 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 432.8$$

$$= 432.800 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[185.671, 432.8, 1136.0] + 142.0$$

$$= 327.671 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 615.6)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min\left(12, \left(2 \cdot R_n + t_n\right) / \left(\sqrt{\left(D_i + t_{eff}\right) \cdot t_{eff}}\right)\right)$$

$$= \min\left(12, \left(2 \cdot 307.8 + 112.0\right) / \left(\sqrt{\left(8004.0 + 142.0\right) \cdot 142.0}\right)\right)$$

$$= 0.677$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right)$$

$$= 142.0 \cdot 615.6 \cdot \max\left(\left(0.677/5\right)^{0.85}, 1\right)$$

$$= 874.152 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B$$

$$= 287.84 + 54.179$$

$$= 342.020 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 874.152 + 0.951(342.02 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 1199.775 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 112.0 / \ln[1 + 112.0/307.8]$$

$$= 360.908 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$

$$= 55.0 \cdot 360.908 \cdot 327.671$$

$$= 66327.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs}(LR + t_n)$$

$$= 55.0 \cdot 4072.584(615.6 + 112.0)$$

= 1661970.1 Kgf

Discontinuity Force from Internal Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
 = $55.0 \cdot 4072.584 \cdot 307.8$
 = 703070.9 Kgf

Area Resisting Internal Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
 = $(1661970 + 703071 + 66327.7) / 55.0$
 = 43350.1 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$
 = $241.65 / (2 \cdot 43350.125 / 1199.775 - 4072.584 / 142.0)$
 = 55.4 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S / [t / R_{xs}]$
 = $161.1 [142.0 / 4072.584]$
 = 56.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
 = $\min(55.442, 56.168)$
 = 55.442 bars

Average Primary Membrane Stress [σ_{Avg}]:

= $(f_N + f_S + f_Y) / AT$
 = $(66327.695 + 1661970 + 703071) / 1199.775$
 = 198.737 N./mm²

General Primary Membrane Stress [σ_{Circ}]:

= $P \cdot R_{xs} / t_{eff}$
 = $55.0 \cdot 4072.584 / 142.0$
 = 157.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= $\max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$
 = $\max(2 \cdot 198.737 - 157.751, 157.751)$
 = 239.7 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	239.72	N./mm ²
Maximum Allowable Working Pressure	Pmax	55.44	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

= $(R_{nc} + t_n) / R_{nc}$
 = $(307.8 + 112.0) / 307.8$
 = 1.364 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

= $\pi / 2 \cdot (R_n + t_n)$
 = $\pi / 2 \cdot (307.8 + 112.0)$
 = 659.420 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 112.000, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot ky, 1.5 \cdot S_n(A_2 + A_3), \pi / 4 \cdot P \cdot R_n^2 \cdot ky^2)$
 = $\min(703071 \cdot 1.36, 1.5 \cdot 153.1(342.02 + 0.0), \pi / 4 \cdot 55.0 \cdot 307.8^2 \cdot 1.36^2)$
 = 77631.008 Kgf



Weld Stress Value [τ]:
= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $77631.008 / (659.42(0.49 \cdot 7.071 + 0.6 \cdot 31.0 + 0.49 \cdot 0.0))$
= 52.324 < or = to 161.100 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT
Computed Minimum Design Metal Temperature : -2 °C
User entered Minimum Design Temperature : 0 °C
The Curve for this Material is : A
From Figure 3.8M- with PWHT, the MDMT is : 8 °C
Temperature reduction per Graph 3.13 : 11 °C
The computed stress ratio is : 0.803

The computed thickness ratio [sr]:
= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(10.434 \cdot 1.0) / (16.0 - 3.0 - 0.0)$
= 0.803

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification : -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -2 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -46 °C

The Drop for this Nozzle is : 22.0901 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 598.8902 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N3

From: 290

Pressure for Reinforcement Calculations	P	55.002	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	161.10	N./mm ²
Parent Allowable Stress At Ambient	Sva	175.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	8000.00	mm.
Shell Finished (Minimum) Thickness	t	144.0000	mm.
Shell Internal Corrosion Allowance	c	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		81499.98	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

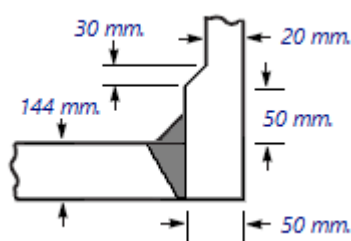
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	153.14	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Hub Height of Integral Nozzle	h	50.0000	mm.
Height of Beveled Transition	L'	30.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	144.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.787	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 80.2 \cdot (\exp(55.002/(153.141 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 1.4665 + 2.0 + 0.0 = 3.4665 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 153.141 \cdot 1.0 \cdot \ln((2 \cdot 18.0 + 80.2) / 80.2) - 0.0$$

$$= 567.798 - 0.0 = 567.7985 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 7.730 mm
 Min. required shell thickness, internal p + c : 141.014 mm.

Nozzle neck minimum required thickness : 7.730 mm.

Minimum nozzle neck thickness 19.990 mm., is greater than required thickness 7.730 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[153.1/161.1, 1]$$

$$= 0.951$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 2.0$$

$$= 48.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 20.0 - 2.0$$

$$= 18.000 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 8004.0/142.0$$

$$= 56.366 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 8000.0/2 + 2.0$$

$$= 4002.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{4002.0 \cdot 142.0}, 2 \cdot 40.1)$$

$$= 80.200 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 142.0, 0.0] + \sqrt{40.1 \cdot 48.0}$$

$$= 43.873 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(142.0 + 0.0)$$

$$= 1136.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[43.873, 250.0, 1136.0] + 142.0$$

$$= 185.873 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 142.0 + ((0.0 \cdot 1.0) / 80.2)$$

$$= 142.000 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 40.1 + 48.0) / (\sqrt{(8004.0 + 142.0) \cdot 142.0}))$$

$$= 0.119$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 142.0 \cdot 80.2 \cdot \max((0.119/5)^{0.85}, 1)$$

$$= 113.884 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 48.0 \cdot 185.873$$

$$= 89.219 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 113.884 + 0.951(89.219 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 199.195 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 48.0 / \ln[1 + 48.0/40.1]$$

$$= 60.984 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 142.0 / \ln[1 + 142.0/4002.0]$$

$$= 4072.584 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 161.1 \cdot 1.0$$

$$= 241.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 55.002 \cdot 60.984 \cdot 185.873$$

$$= 6357.7 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 55.002 \cdot 4072.584(80.2 + 48.0)$$

$$= 292839.5 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 55.002 \cdot 4072.584 \cdot 40.1$$

$$= 91598.0 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (292839.5 + 91598.0 + 6357.7) / 55.002$$

$$= 6967.5 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:
 = Sallow / (2 • Ap/AT - Rxs/teff)
 = 241.65 / (2 • 6967.511/199.195 - 4072.584/142.0)
 = 58.5 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = S[t/Rxs]
 = 161.1[142.0/4072.584]
 = 56.2 bars

Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(58.541, 56.168)
 = 56.168 bars

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (6357.703 + 292839.5 + 91598.008)/199.195
 = 192.397 N./mm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / teff
 = 55.002 • 4072.584/142.0
 = 157.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 192.397 - 157.755, 157.755)
 = 227.0 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	241.65	N./mm ²
Local Primary Membrane Stress	PL	227.04	N./mm ²
Maximum Allowable Working Pressure	Pmax	56.17	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (40.1 + 48.0)/40.1
 = 2.197 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:
 = π/2 • (Rn + tn)
 = π/2 • (40.1 + 48.0)
 = 138.387 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be >= min(0.7 • 48.000, 6.000) Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), π/4•P•Rn²•ky²)
 = min(91598•2.2, 1.5•153.1(89.219+0.0), π/4•55.0•40.1²•2.2²)
 = 3419.112 Kgf

Weld Stress Value [τ]:
 = fwelds/(Lt(0.49•L41T + 0.6•tw1 + 0.49•L43T))
 = 3419.112/(138.387(0.49•7.071 + 0.6•142.0 + 0.49•0.0))
 = 2.733 < or = to 161.100 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 0.9705 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 394.9706 mm.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: